*fakeWind Seminar, OSU, 5/1/2020*

# Hybrid Simulation: Overview, Theory, and Applications

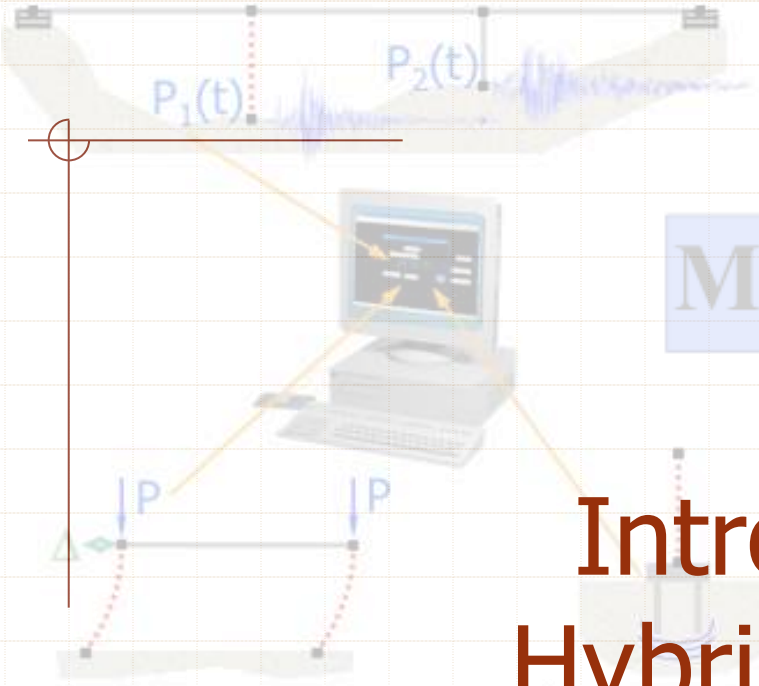
Andreas Schellenberg, Ph.D., P.E.
*Associate Principal, Maffei Structural Engineering*

MAFFEI
STRUCTURAL ENGINEERING



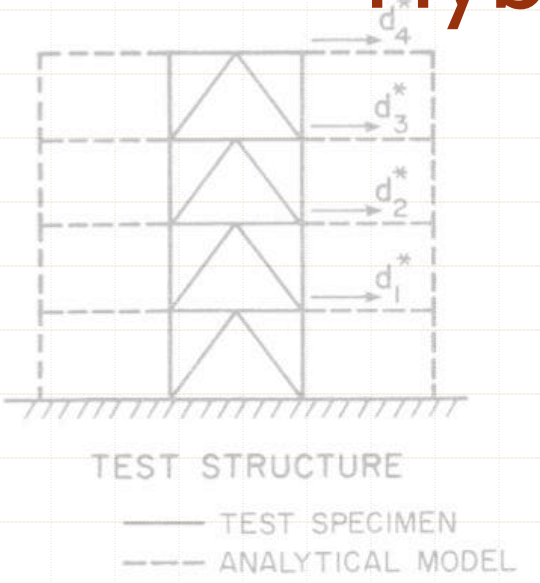
# Outline of Presentation

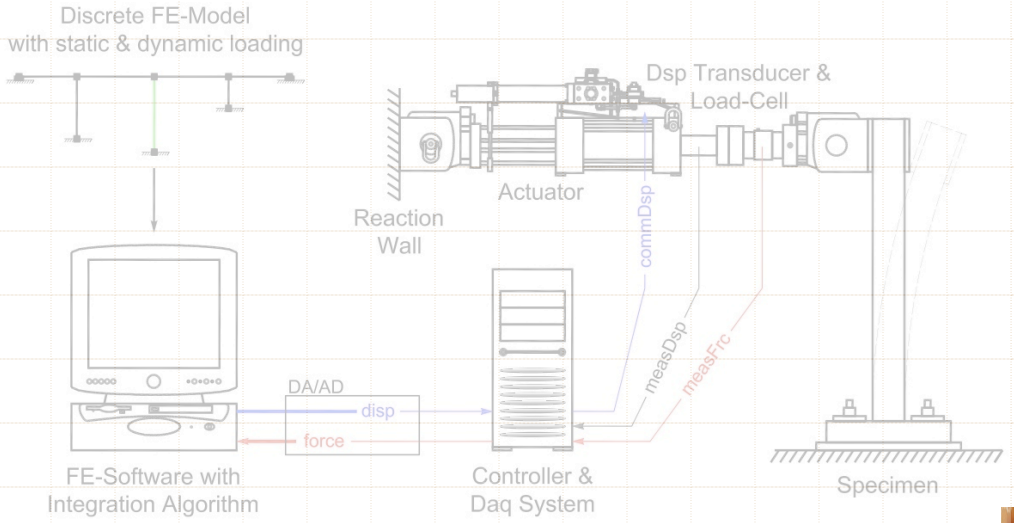
1. Introduction to Hybrid Simulation
2. OpenFresco Architecture
3. Other OpenFresco Features
4. Implementation – Things to look out for
5. Test Accuracy and Assessment
6. Summary & Conclusions




$$\mathbf{M}\cdot\ddot{\mathbf{u}}+\mathbf{C}\cdot\dot{\mathbf{u}}+\mathbf{P}_r(\mathbf{u})=\mathbf{P}(t)$$

# Introduction to Hybrid Simulation



# Cyber-Physical Systems (CPS)

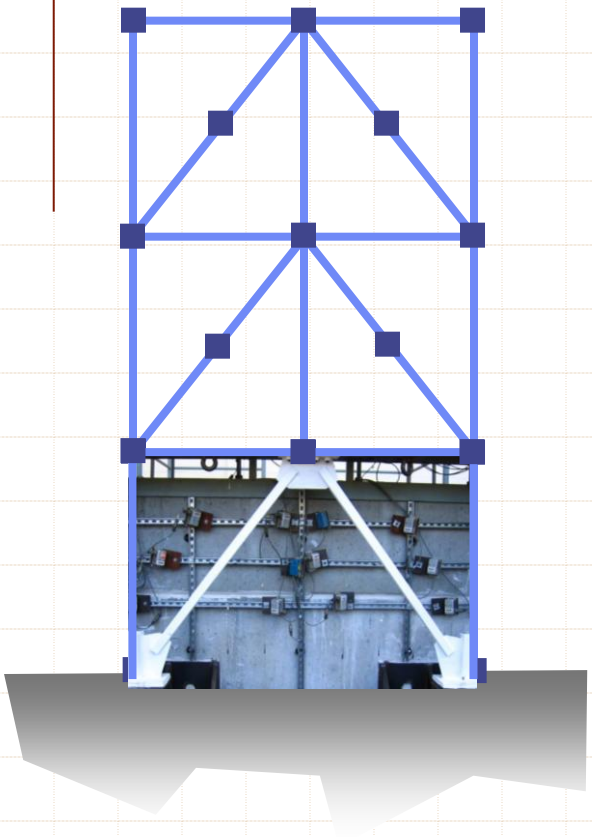
- A system where *physical and software components are deeply intertwined, each operating on different spatial and temporal scales, exhibiting multiple and distinct behavioral modalities, and interacting with each other in a myriad of ways that change with context* (from US NSF)
- Hybrid Simulation (HS), a flexible and economical experimental testing method, can be considered a cyber-physical system
- Control system aspects play a central role in CPS as well as HS

# Comparison of Exp. Test Methods

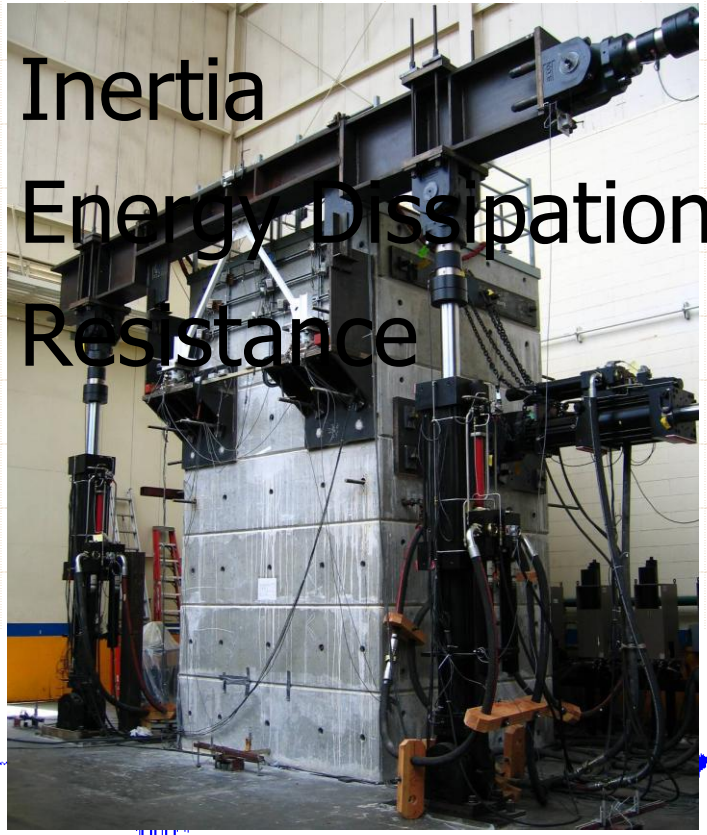
| | Quasi-Static | Wave Basin | Hybrid Simulation |
|---|---|---|---|
| **Dynamics** | NO | YES | YES |
| **Real-Time Effects** | NO | YES | YES (if real-time test) |
| **Large- or Full-Scale** | YES | NO (limited by basin) | YES |
| **Wind Loads** | NO | NO | YES |

# Structural Hybrid Simulation

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{P}_r(\mathbf{u}) = \mathbf{P}(t)$$

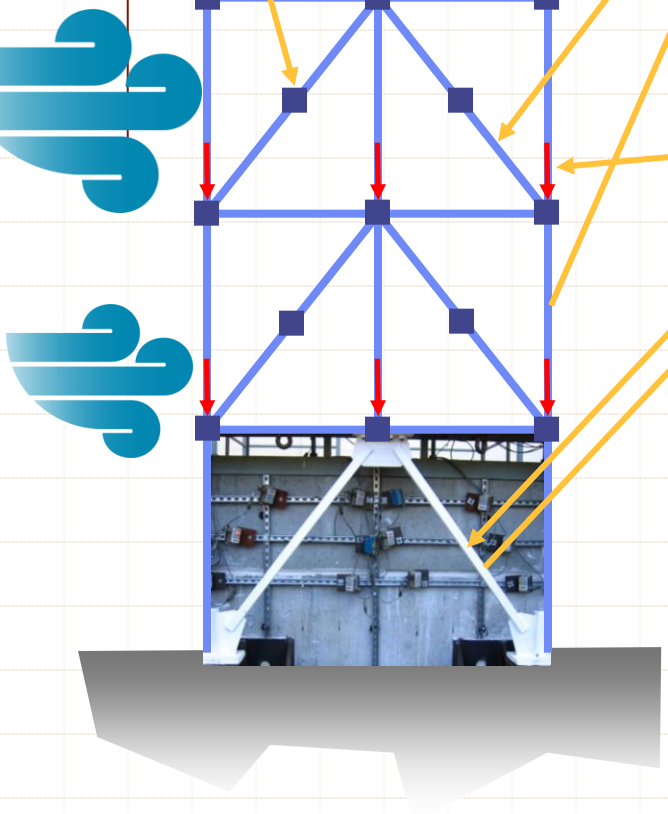


- Inertia
- Energy Dissipation
- Resistance



# Aero Hybrid Simulation

$$M\ddot{x} + C\dot{x} + P_r(u) = P(t)$$



Dynamic Loading:
- Seismic
- Wind
- Blast/Impact
- Wave
- Traffic

Static Loading:
- Gravity
- Prestress

analytically add nonlinear geometric effects to measured resisting forces

analytical model of structural energy dissipation and inertia

physical model of structural resistance

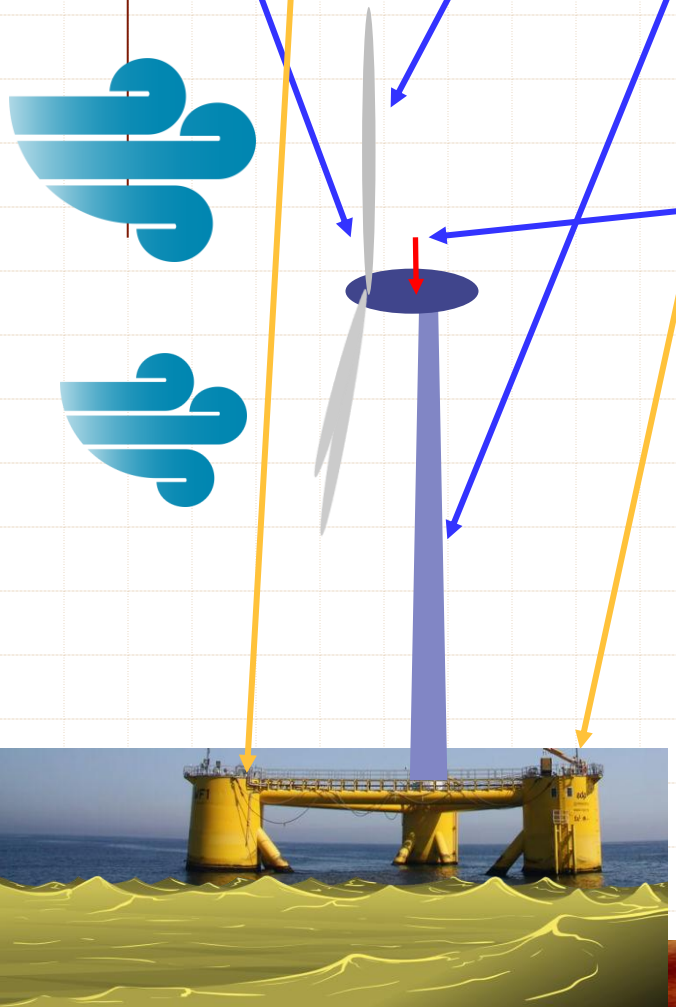
# Aero-Hydro-RTHS

$$M\ddot{u} + C\dot{u} + P_r(u) = P(t)$$

Dynamic Loading:
- Seismic
- Wind
- Blast/Impact
- Wave
- Traffic

Static Loading:
- Gravity
- Prestress

Turbine Control:



analytical model of structural tower, turbine, rotor, and aerodynamic loads

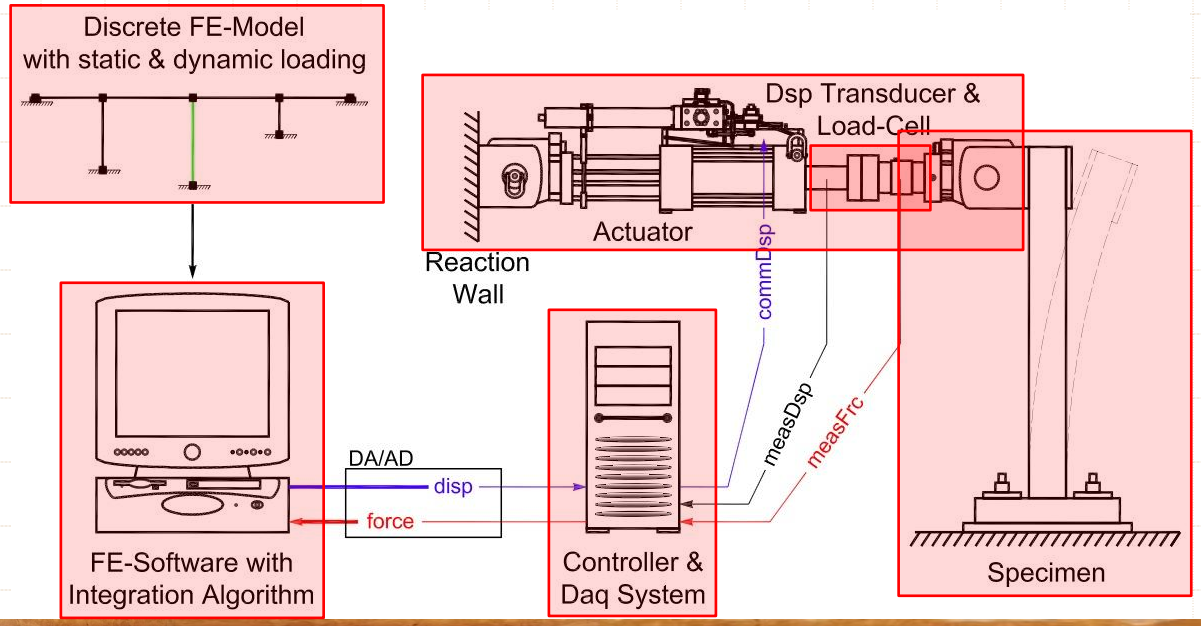
physical model of floating platform, moorings, and hydrodynamic loads

# Hybrid Simulation

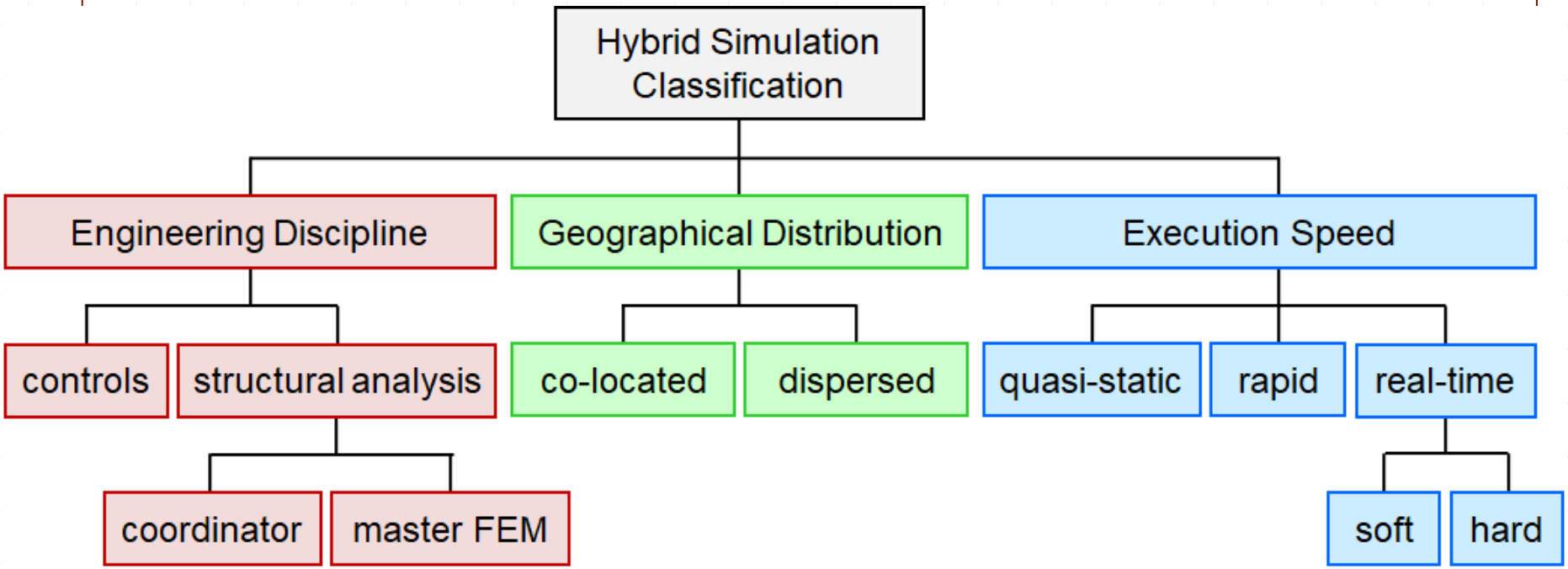
- Model the well understood parts of a structure and/or loading conditions in a finite element program on one or more computers (HPC possible)
- Leave the construction and testing of the highly nonlinear and/or numerically hard to model parts of the structure and/or loading conditions in one or more laboratories
- Can be considered as a conventional finite element analysis where physical models of some portions of the structure are embedded in the numerical model

# Required Components

1. Discrete model of the structure to be analyzed, including the static and aerodynamic loading
2. Control system with dynamic actuation / transfer system and reaction frame / support structure
3. Physical test specimen incl. hydrodynamic loading
4. Data acquisition system with instrumentation



# Hybrid Simulation Classification



# Advantages

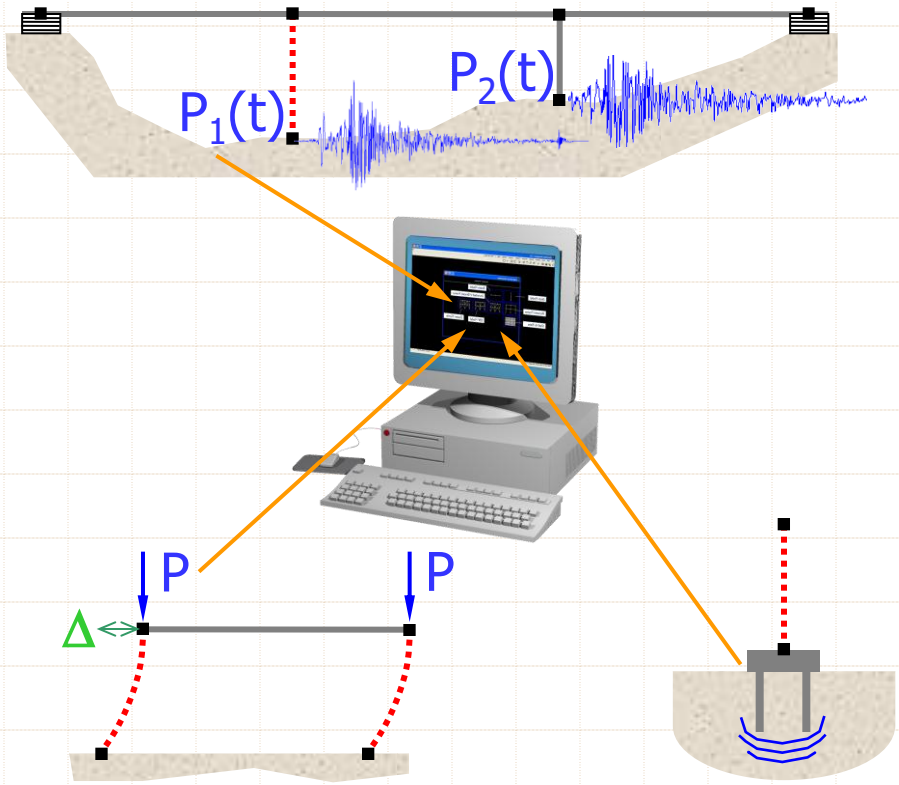
- Enables dynamic testing of full-scale specimens
- Quasi-static testing equipment sufficient
- Fewer restrictions on size, weight and strength of a specimen

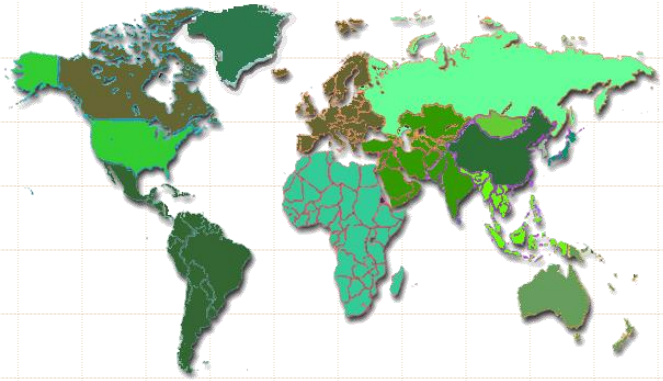





# Advantages

- Geometric nonlinearities, three-dimensional effects, multi-support excitations and soil-structure interactions can be incorporated into the analytical model

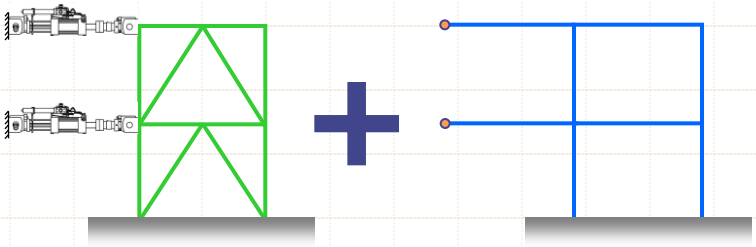


- Internationally, geographically distributed testing is made possible

# Advantages

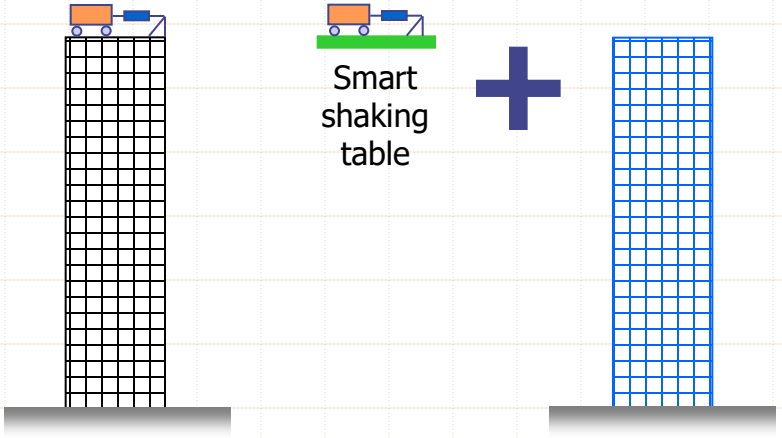
## Quasi-static applications



Drifts may concentrate in one story

Numerical model of portions of system that may alleviate weak story response

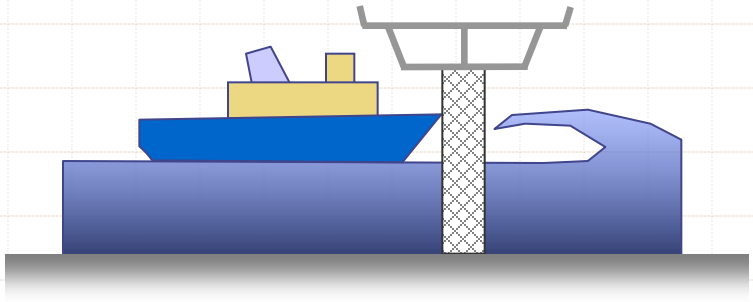
## Dynamic Applications




Shaking Table

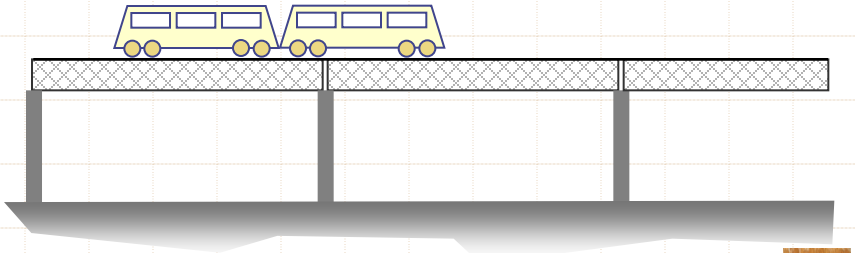
Numerical Model

## Hydrodynamic effects



Including impact

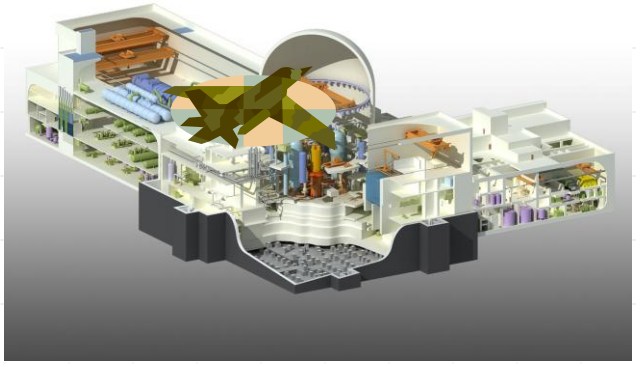
## Moving Loads



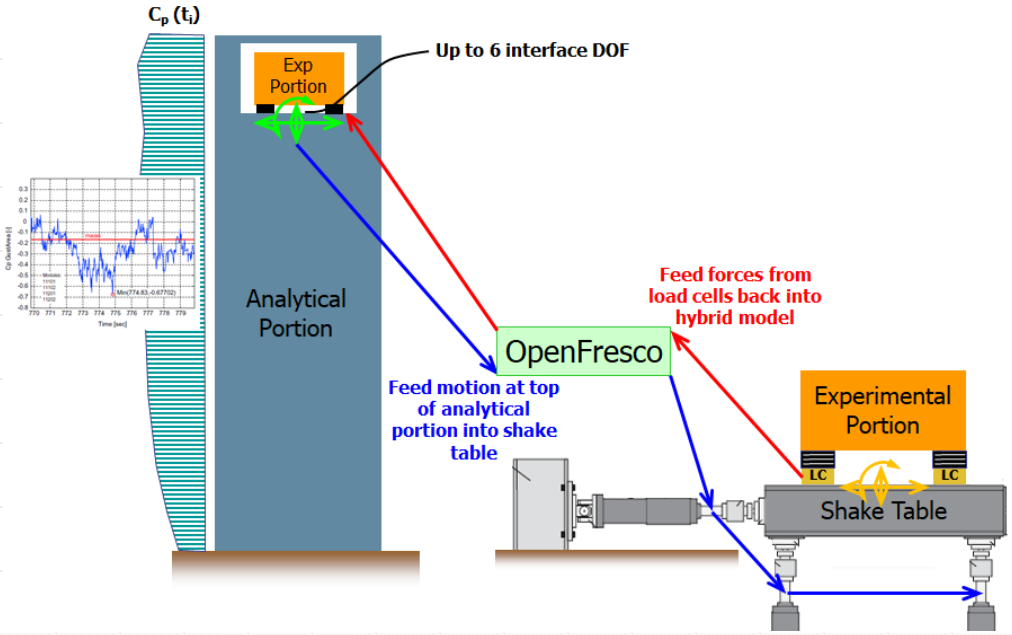
Interaction with moving loads, including impact

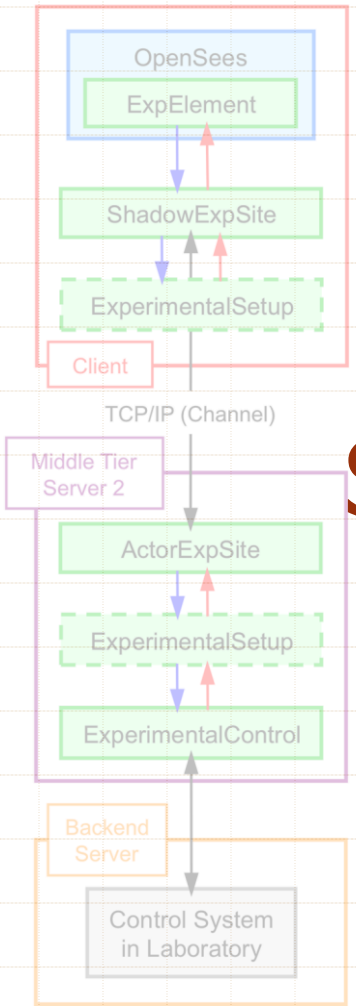
# Advantages

- Aircraft Impact



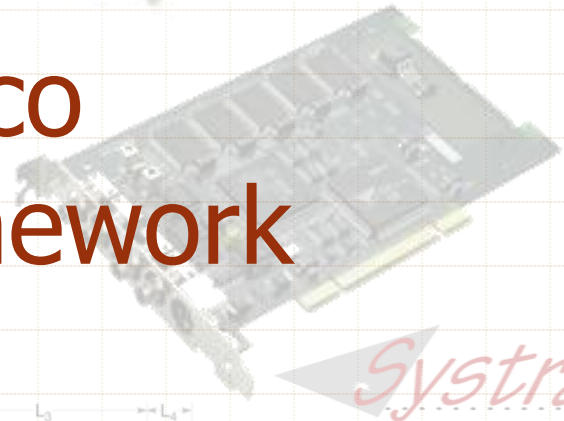
- Aerodynamic Loads + Struct. HS

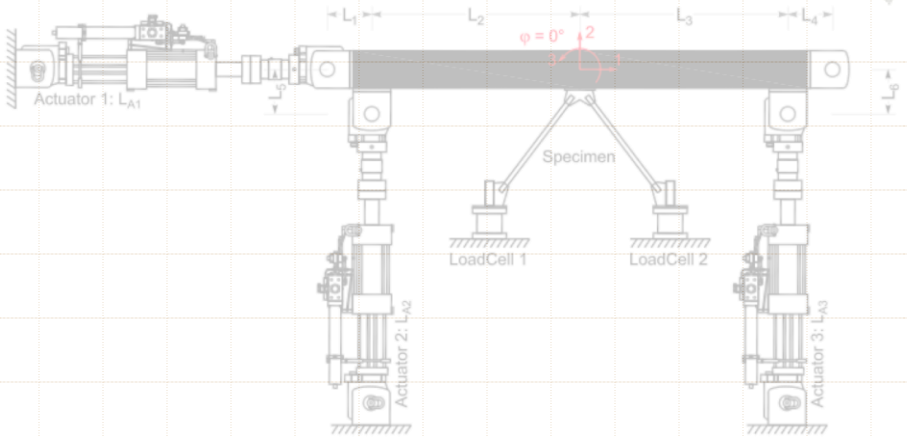







# OpenFresco Software Framework





# What is OpenFresco?

- Open source Framework for Experimental Setup and Control
- Secure, object oriented, network enabled "middleware" -- Pairs computer analysis software with laboratory control systems and other software to enable hybrid and collaborative computing:

- Computational Drivers
  - OpenSees
  - OpenFresco*Express*
  - Abaqus
  - LS-DYNA
  - Matlab
  - Simulink
  - Ansys
  - UI-SimCor
  - OpenFAST
  - OpenSees PFEM
- Control Systems
  - dSpace
  - MTS
    - STS family
    - Flextest/CSI
    - Flextest/SCRAMNet
    - 469D
    - SRMD
  - National Instruments
  - Pacific Instruments
  - ADwin

# Why a Software Framework?

- Lack of a common framework for development and deployment of HS
- Problem specific implementations which are site and control system dependant
- Such highly customized software implementations are difficult to adapt to different structural problems

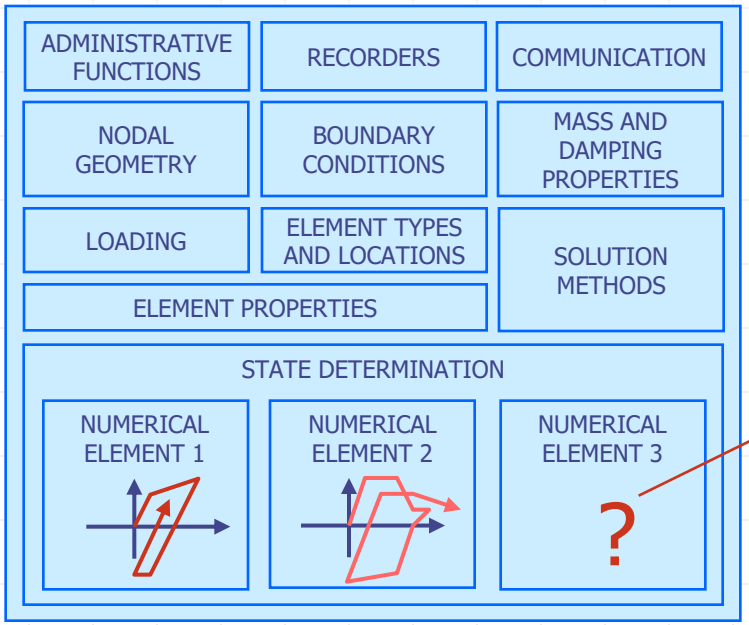
→ Need a robust, transparent, adaptable, and easily extensible software framework for research and deployment

# What is a Software Framework?

- A reusable design for a software system, or subsystem
- Defines overall architecture of a software system, meaning its basic components and the relationships among them
- Expressed as a set of abstract classes and the way their instances collaborate
- Loose-coupling of components within the framework is essential for extensibility and reusability

# Rethinking implementation strategies

- Embed test specimen(s) in an existing computational framework of user's choice

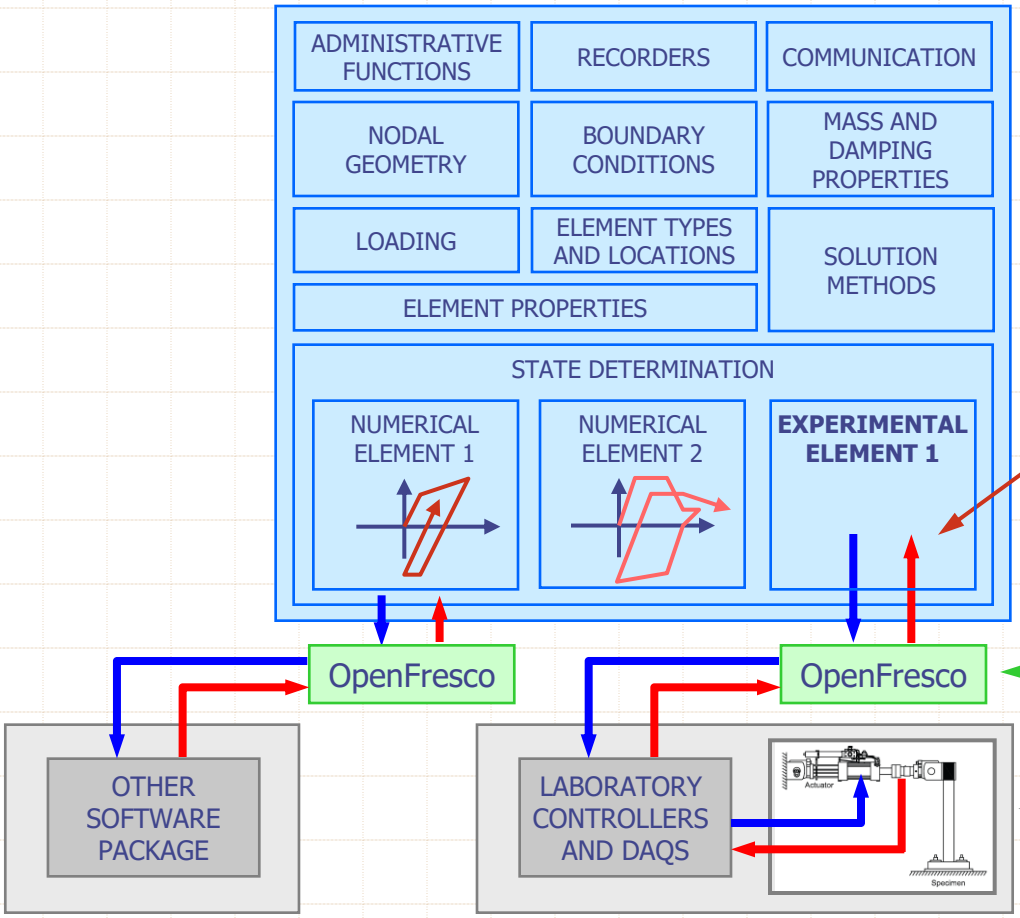



Typical features of an analysis framework

Proper numerical model uncertain

# Rethinking implementation strategies

- Embed test specimen(s) in an existing computational framework of user's choice



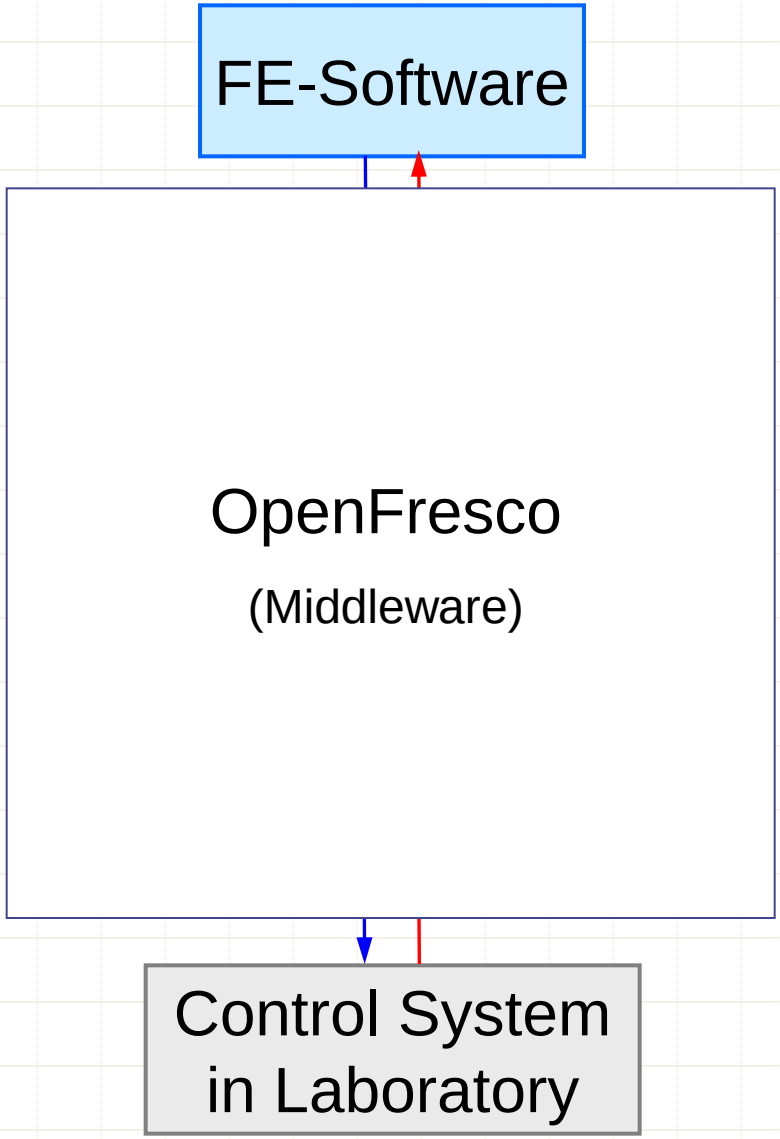

Typical features of an analysis framework

Define element as an "Experimental Element"

OpenFresco

Laboratory

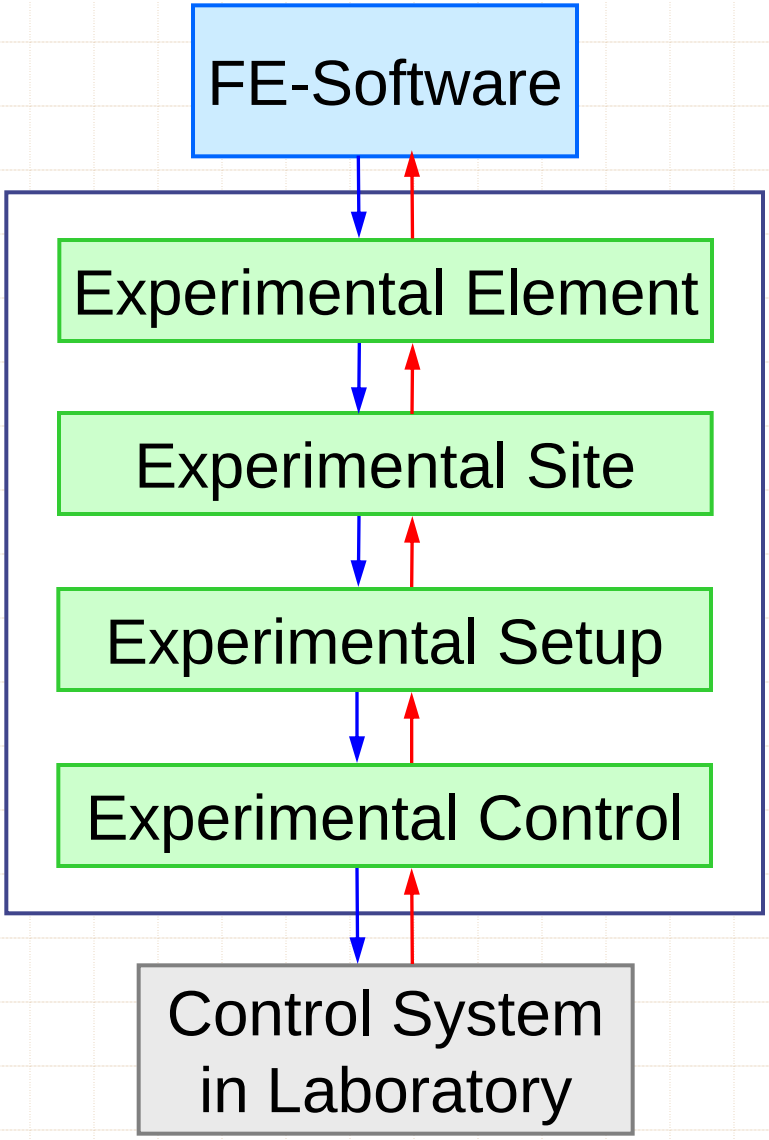
# OpenFresco Components




provides all features of unmodified computational framework, including parallel and network computing

provides control of physical actuators as well as data acquisition using physical instrumentation devices

# OpenFresco Components




provides all features of unmodified computational framework, including parallel and network computing

represents the part of the structure that is physically tested and provides the interface between the FE-software and the experimental software framework

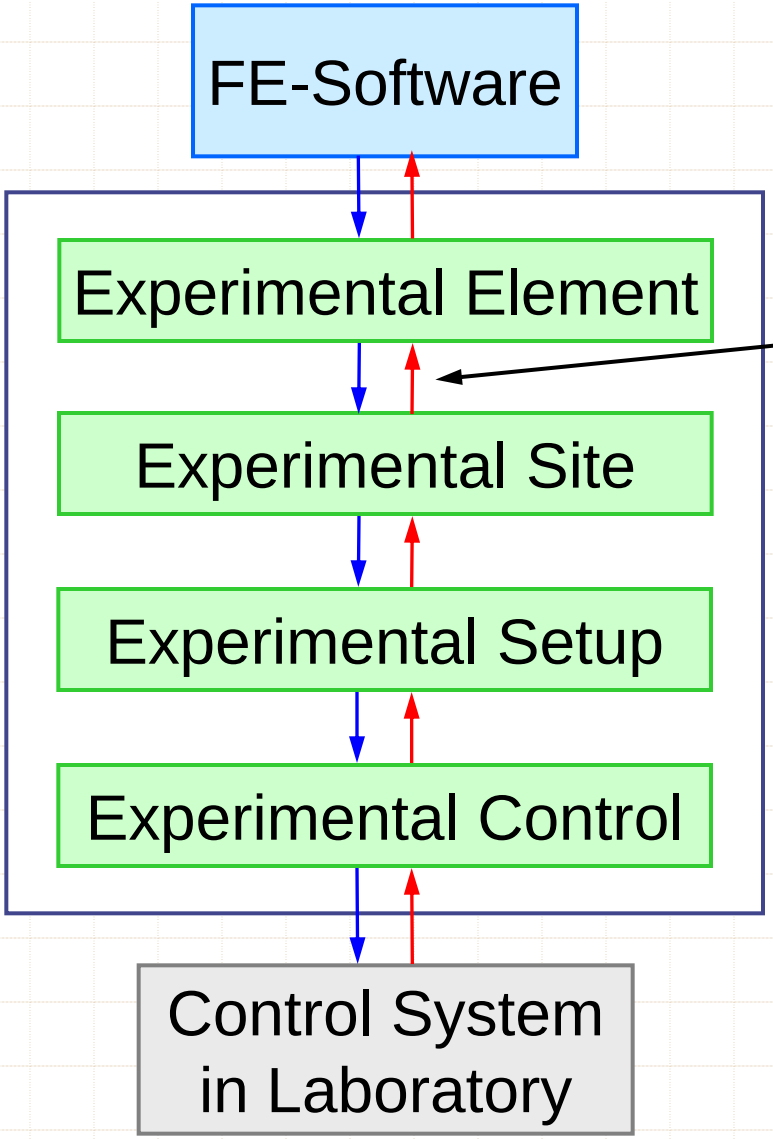
stores data and provides communication methods for distributed testing

transforms between the experimental element degrees of freedom and the actuator degrees of freedom (linear or non-linear transformations)

interfaces to the different control and data acquisition systems in the laboratories

provides control of physical actuators as well as data acquisition using physical instrumentation devices

# OpenFresco Components




provides all features of unmodified computational framework, including parallel and network computing

Signal variables that currently can be transferred:

- Displacements
- Velocities
- Accelerations
- Forces
- Time

provides control of physical actuators as well as data acquisition using physical instrumentation devices

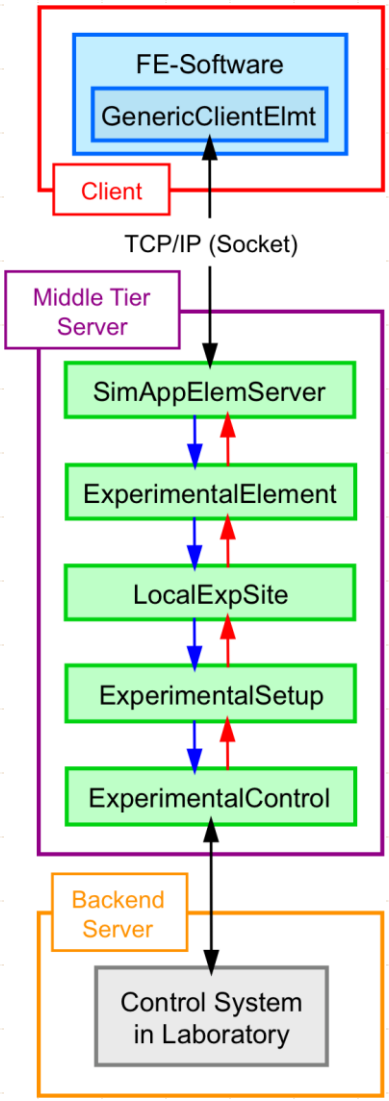
# Requirements for Architecture

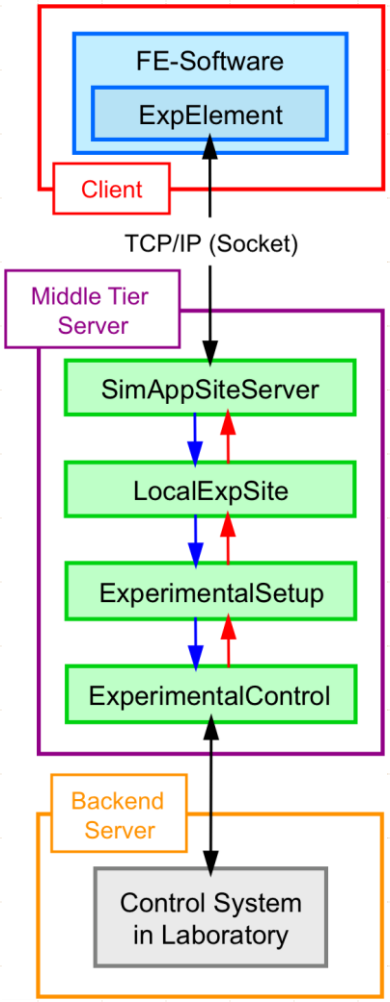
- Provide connectivity to a wide variety of FE-software (clients), independent of the language, such analysis software is programmed in
- Enable distributed testing and support different communication protocols
- Interface with rapidly evolving control and data acquisition systems deployed at testing facilities all over the world

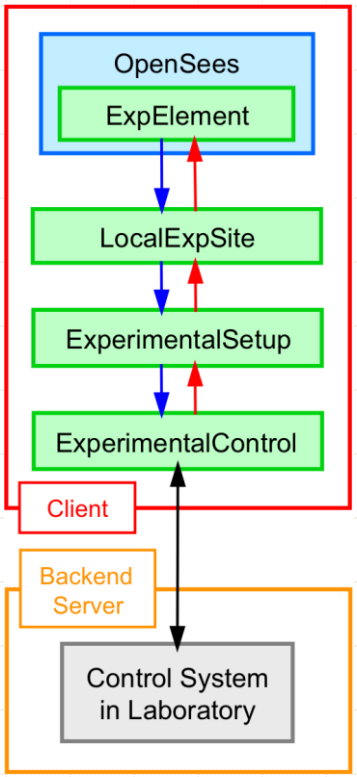
→ Multi/Three-Tier Software Architecture

# OpenFresco Components

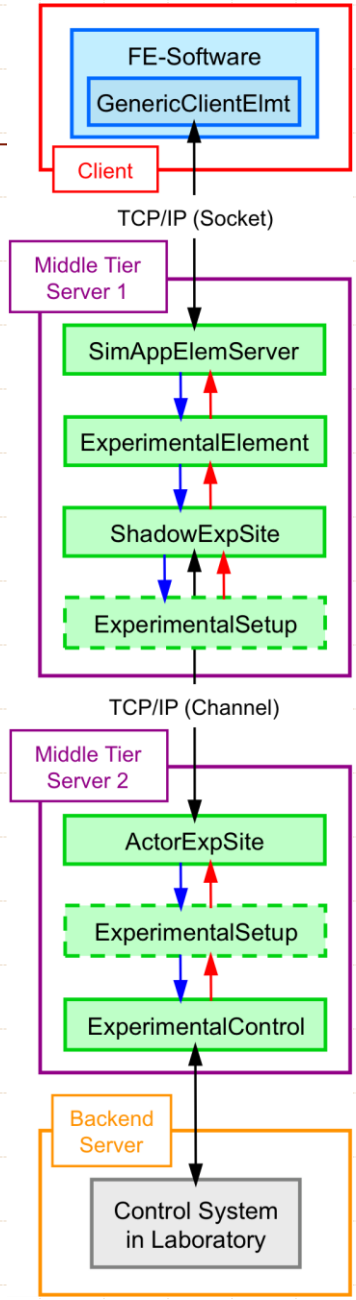
local deployment

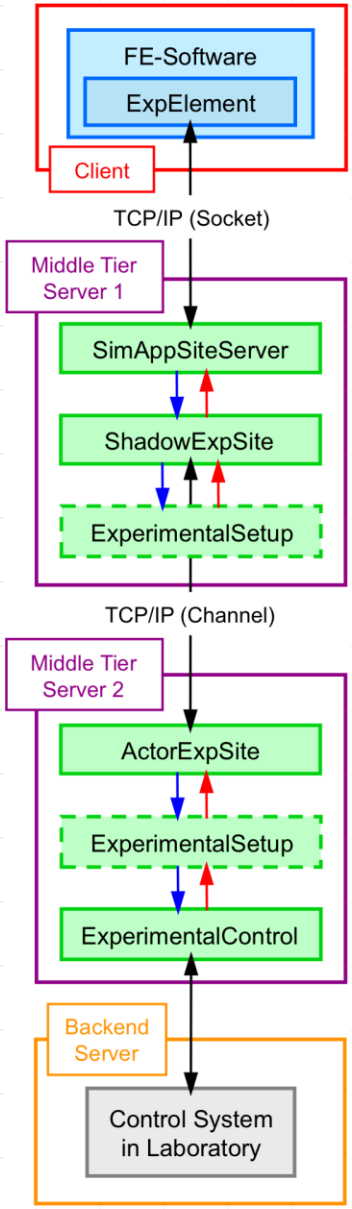


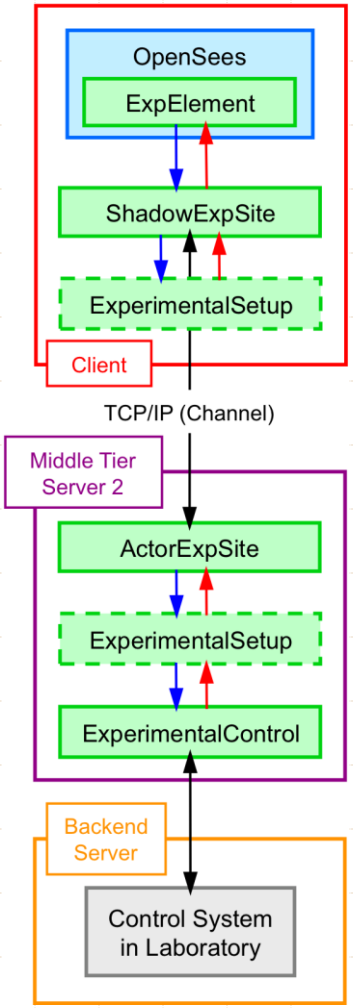



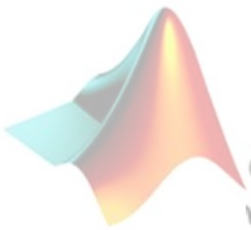



# network deployment





MATLAB®
SIMULINK®

OpenSees

# Computational Drivers

ABAQUS



LSTC
Livermore Software
Technology Corp.

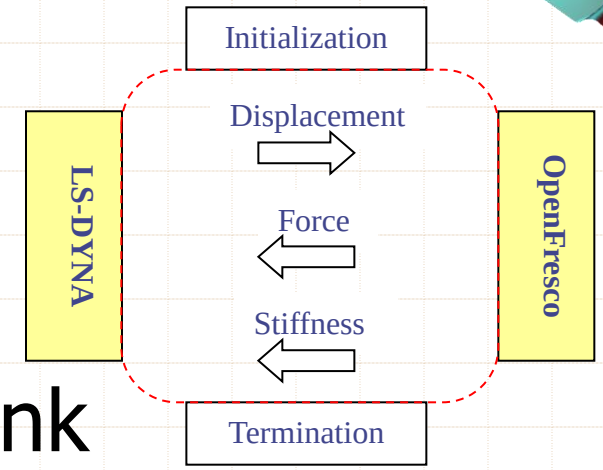
ANSYS®

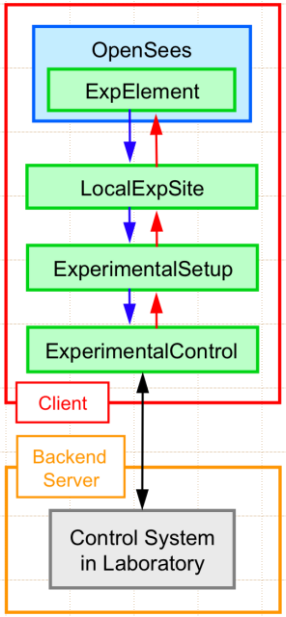
# How to Interface

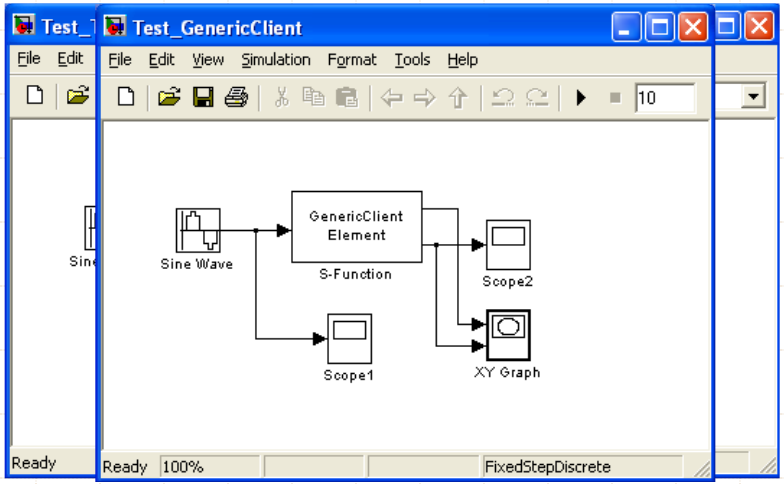
- Two Ways to Interface with FE-Software
  - Generic Client Element
  - Experimental Element Directly in FE-Software
- Generic Client Element to be Programmed by the Developers
- Several generic client elements available: /trunk/SRC/simApplicationClient

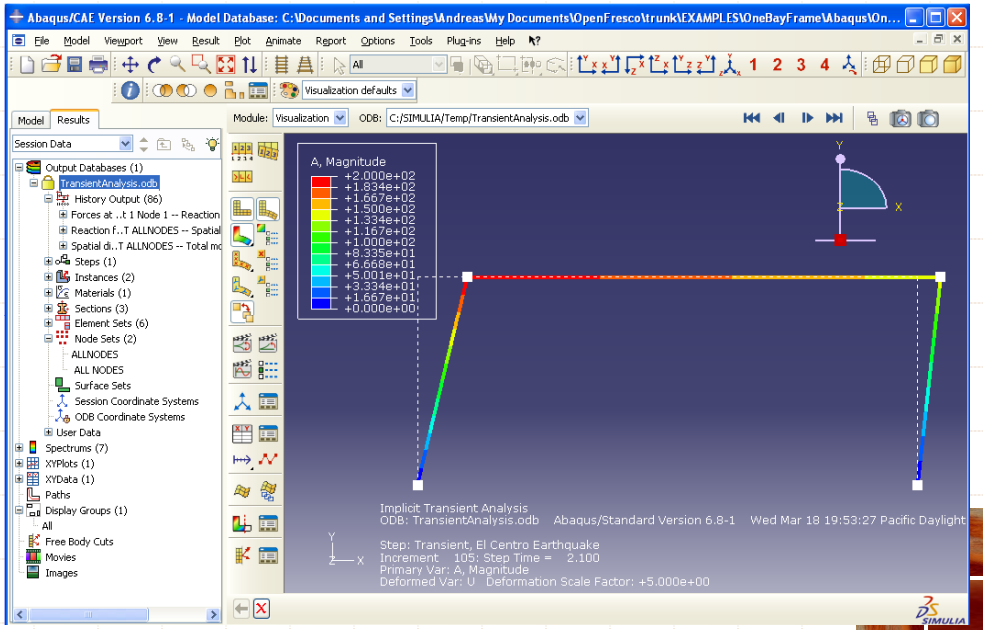
# Computational Drivers

- OpenSees
- LS-DYNA
- Abaqus
- Matlab/Simulink
- UI-SimCor

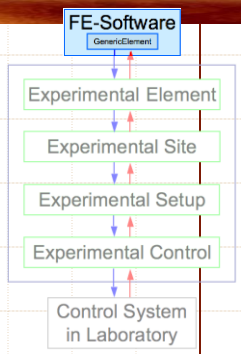


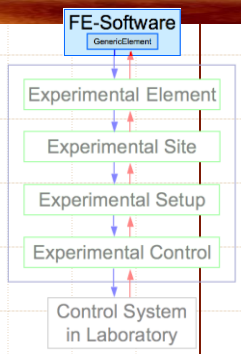
# Time Integration Methods




$$\mathbf{M}\ddot{\mathbf{u}}_n + \mathbf{C}\dot{\mathbf{u}}_n + \mathbf{P}_r(\mathbf{u}_n) = \mathbf{P}(t_n)$$

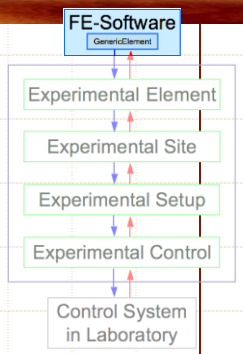
- Mass matrix M is often singular
  -> second order differential equation
  infinitely stiff -> fully implicit numerical methods
- Make as few function calls as possible
- Use constant Jacobian in the numerical methods since tangent stiffness is not available

# Direct Integration Methods

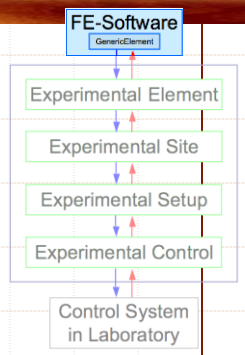



- Explicit Integrators
  - explicit Newmark Method
  - Central-Difference Method
  - explicit Alpha Method
  - explicit Generalized-Alpha Method
  - KR-Alpha Method
- Implicit Integrators (do not use for HS)
  - Newmark Method
  - Alpha Method
  - Generalized-Alpha Method
  - Collocation Method

# Direct Integration Methods

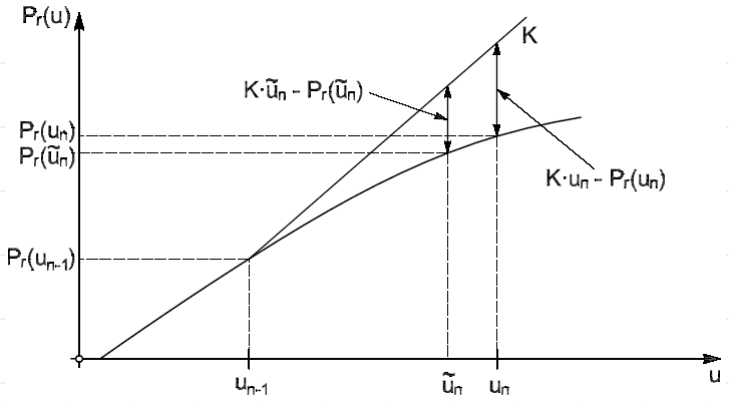


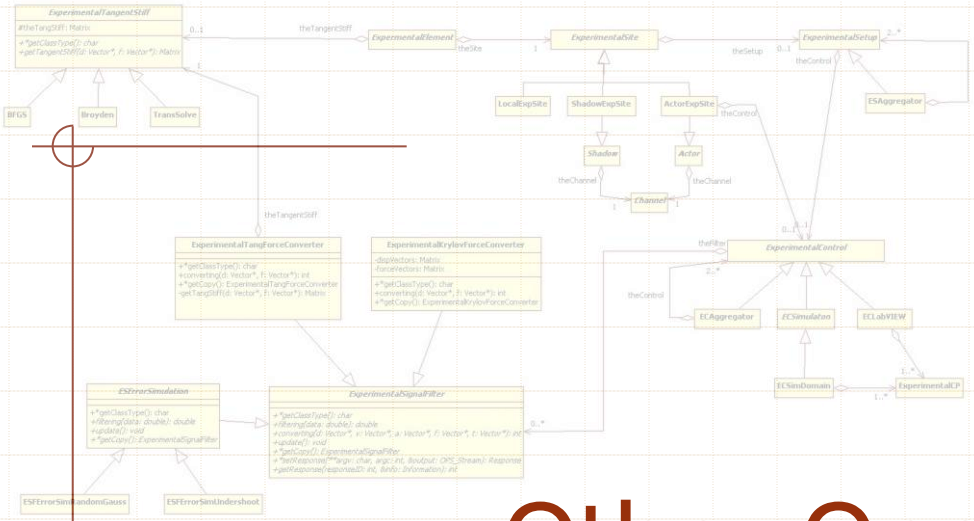

- Implicit Integrators with increment reduction factors
  - Newmark HS IncrReduct Method
  - Generalized-Alpha HS IncrReduct Method
  - Collocation HS IncrReduct Method
- Implicit Integrators with increment limits
  - Newmark HS IncrLimit Method
  - Generalized-Alpha HS IncrLimit Method
  - Collocation HS IncrLimit Method

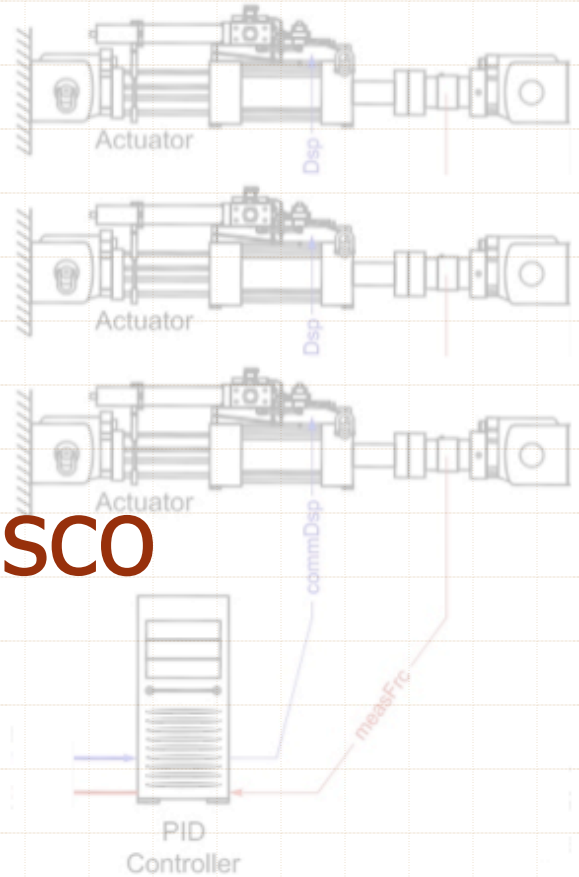



# Direct Integration Methods

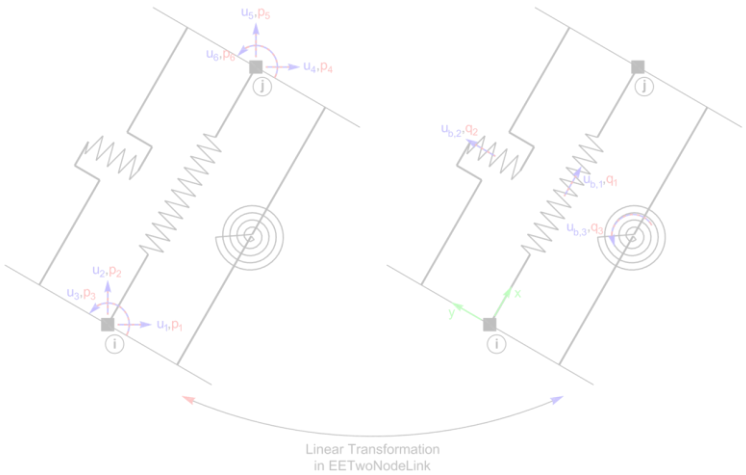
- Implicit Integrators with sub-stepping (constant number)
  - Newmark HS FixedNumIter Method
  - Generalized-Alpha HS FixedNumIter Method
  - Collocation HS FixedNumIter Method
- Predictor-Corrector Integrators
  - Alpha-OS Method
  - Generalized-Alpha-OS Method






# Other OpenFresco Features



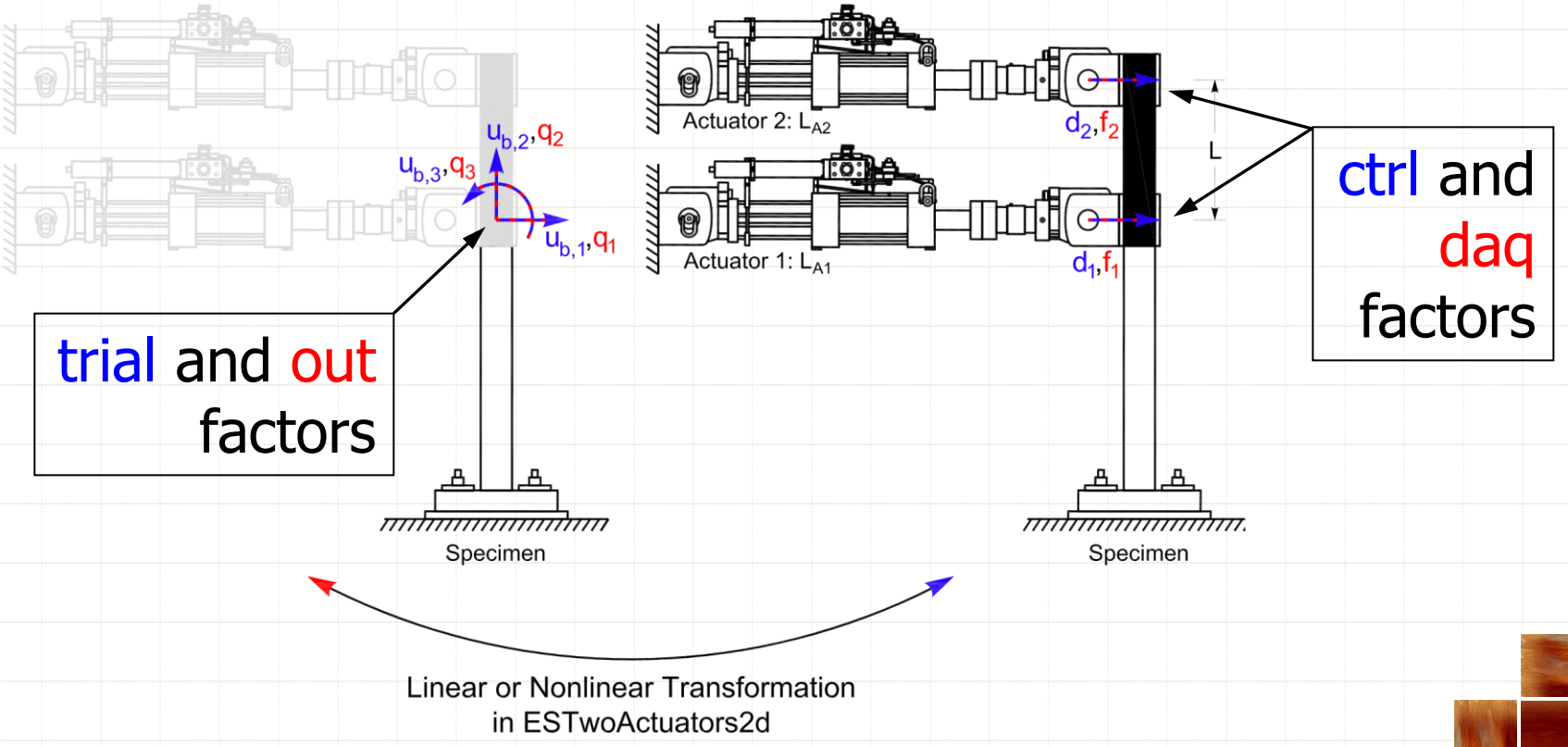

$$u_{b,1} = u_4 - u_1$$
$$u_{b,2} = u_5 - u_2$$
$$-0.5L(u_3 + u_6)$$
$$u_{b,3} = u_6 - u_3$$



# Similitude and Scaling

- Response Factors:

  factors can now be applied to trial and output response data in addition to control and daq response data

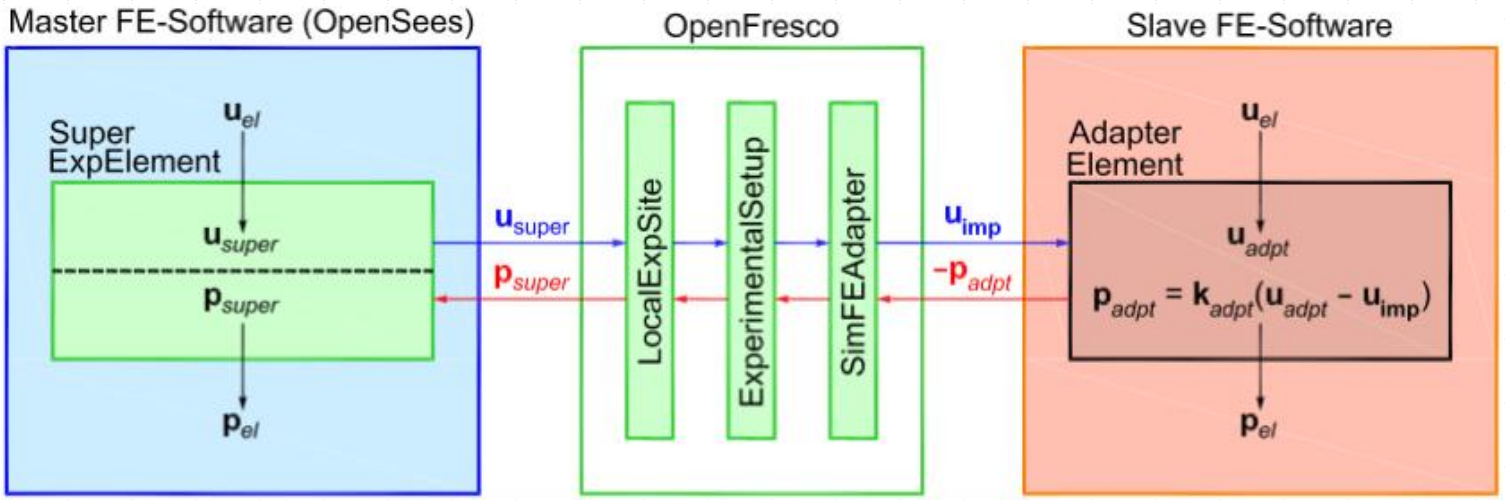


# Co-Simulation (Software Coupling)

## ECSimFEAdapter

```
expControl SimFEAdapter $tag ipAddr $ipPort
```

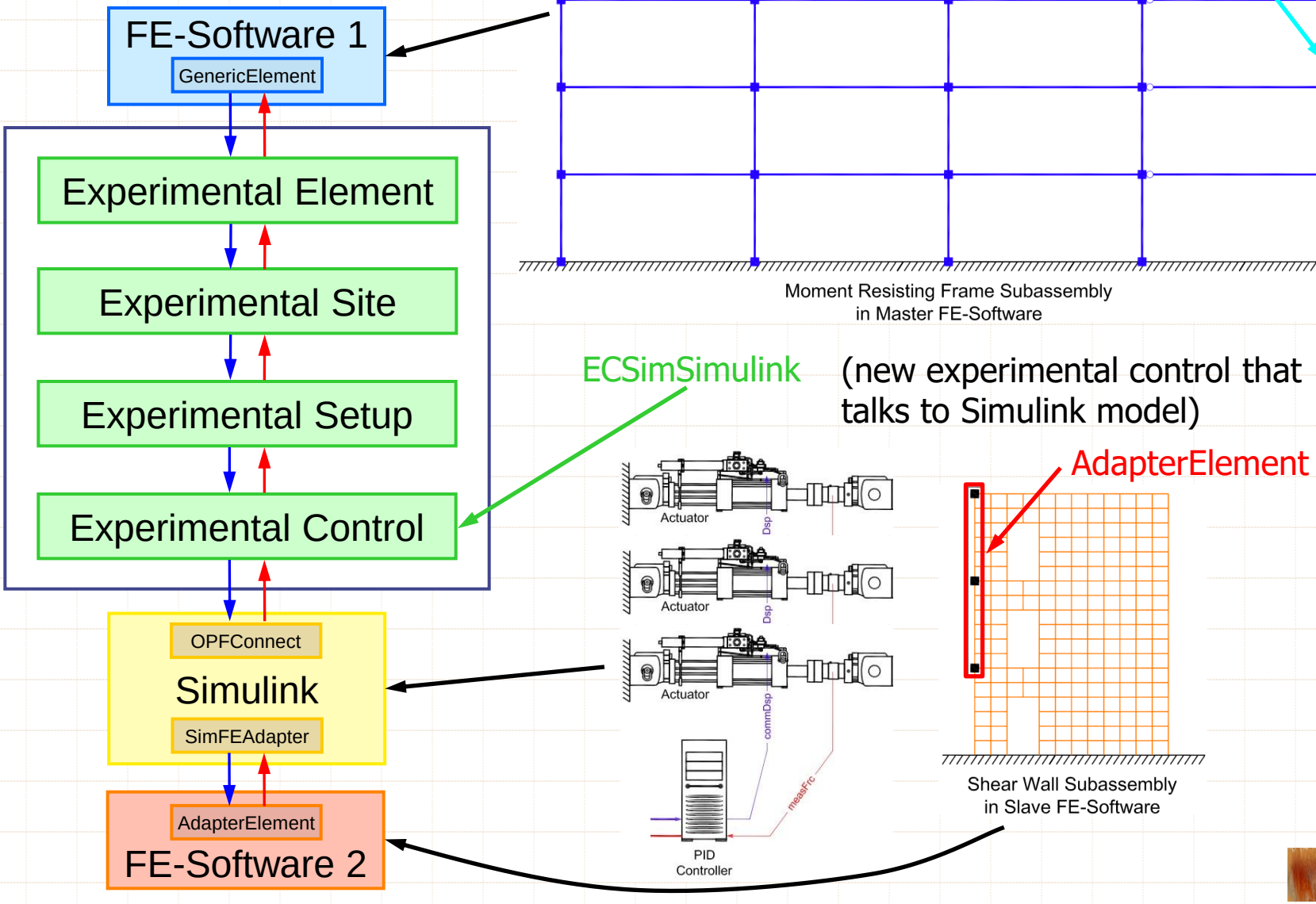
| | |
|---|---|
| `$tag` | unique control tag |
| `ipAddr` | IP address of slave process |
| `$ipPort` | IP port of slave process |



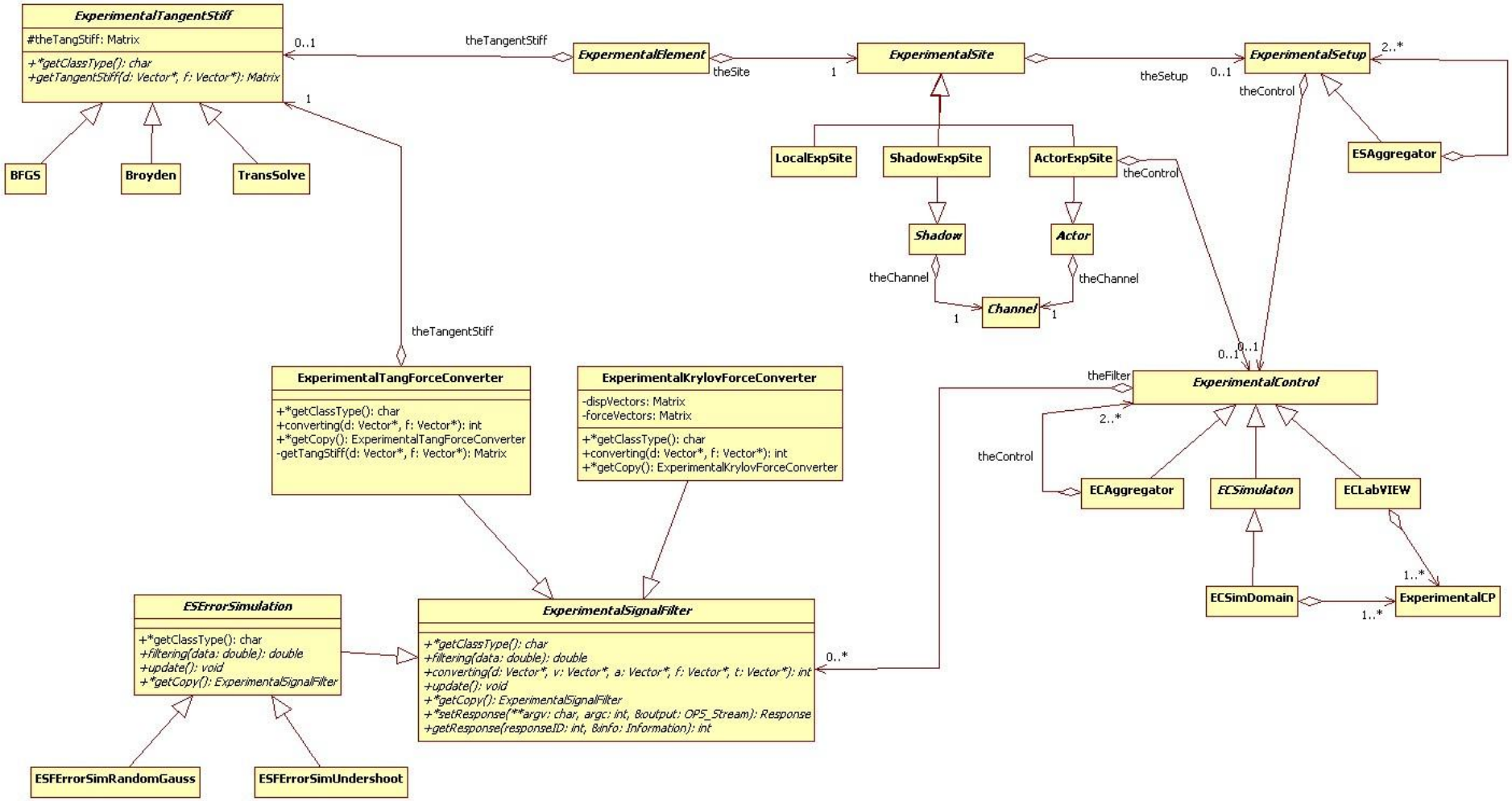
# Co-Simulation (Software Coupling)

- Master Programs
  - OpenSees
  - OpenFresco*Express*
  - Abaqus
  - LS-DYNA
  - Matlab
  - Simulink
  - Ansys
  - UI-SimCor
- Slave Programs
  - OpenSees
  - Abaqus
  - LS-DYNA
  - Matlab
  - Simulink
  - Ansys

# Test Rehearsal

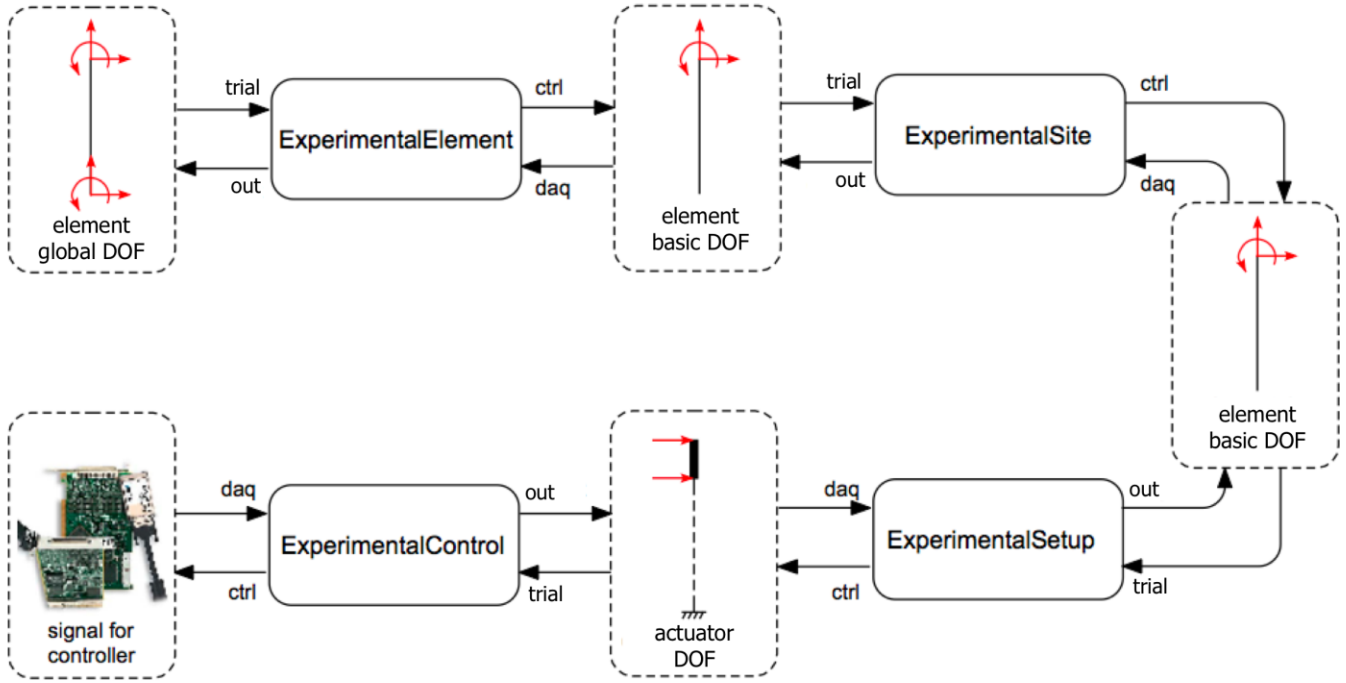


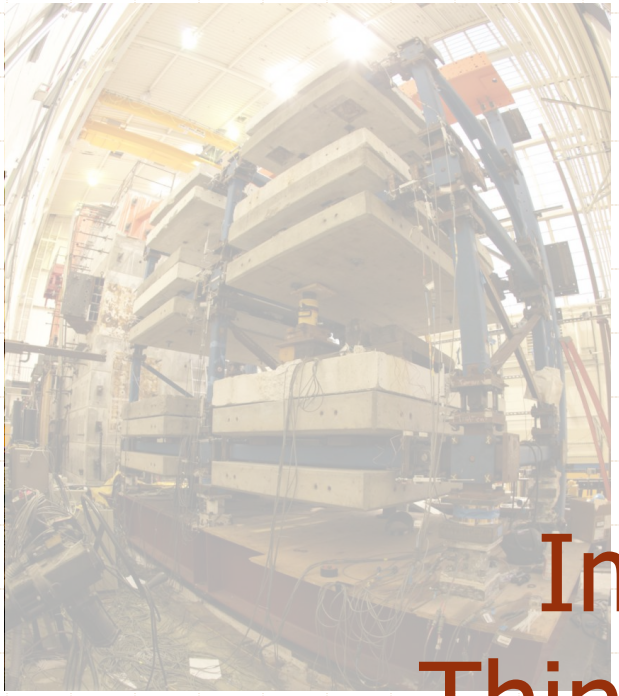
# Force and Mixed Control



# Experimental Recorder

- The recorder objects provide means for monitoring and recording response quantities of interest to the experimentalist.
- Can implement error monitors as recorders.







# Implementation: Things to look out for



One 6DOF Shake Table



# Similitude Laws (for BLWT scaling)

Similitude laws for fluid–elastic models with Froude number neglected.

| Physical quantity | Dimension | Scaling factor with $S_l$, $S_v$, $S_\rho$ | Scaling factor with $S_l$, $S_v$, $S_\rho=1$ | Scaling factor with $S_l=1/30$, $S_v=0.364$, $S_\rho=1$ |
|---|---|---|---|---|
| Length, $l$ | $L$ | $S_l$ | $S_l$ | 0.033 |
| Displacement, $d$ | $L$ | $S_l$ | $S_l$ | 0.033 |
| Velocity, $v$ | $LT^{-1}$ | $S_v$ | $S_v$ | 0.364 |
| Acceleration, $a$ | $LT^{-2}$ | $S_l^{-1}S_v^2$ | $S_l^{-1}S_v^2$ | 3.978 |
| Force, $F$ | $F$ | $S_l^2S_v^2S_\rho$ | $S_l^2S_v^2$ | 0.000147 |
| Time, $t$ | $T$ | $S_lS_v^{-1}$ | $S_lS_v^{-1}$ | 0.092 |
| Modulus, $E$ | $FL^{-2}$ | $S_v^2S_\rho$ | $S_v^2$ | 0.133 |
| Pressure, $p$ | $FL^{-2}$ | $S_v^2S_\rho$ | $S_v^2$ | 0.133 |
| Pressure Coeff, $C_p$ | 1 | 1 | 1 | 1 |
| Stress, $\sigma$ | $FL^{-2}$ | $S_v^2S_\rho$ | $S_v^2$ | 0.133 |
| Strain, $\varepsilon$ | 1 | 1 | 1 | 1 |
| Strain-Rate, $\dot{\varepsilon}$ | $T^{-1}$ | $S_l^{-1}S_v$ | $S_l^{-1}S_v$ | 10.924 |
| Density, $\rho$ | $FL^{-4}T^2$ | $S_\rho$ | 1 | 1 |
| Mass, m | $FL^{-1}T^2$ | $S_l^3S_\rho$ | $S_l^3$ | 0.000037 |
| Damping, c | $FL^{-1}T$ | $S_l^2S_vS_\rho$ | $S_l^2S_v$ | 0.000405 |
| Stiffness, k | $FL^{-1}$ | $S_lS_v^2S_\rho$ | $S_lS_v^2$ | 0.004420 |
| Period, $T$ | $T$ | $S_lS_v^{-1}$ | $S_lS_v^{-1}$ | 0.092 |
| Frequency, $f$ | $T^{-1}$ | $S_l^{-1}S_v$ | $S_l^{-1}S_v$ | 10.924 |

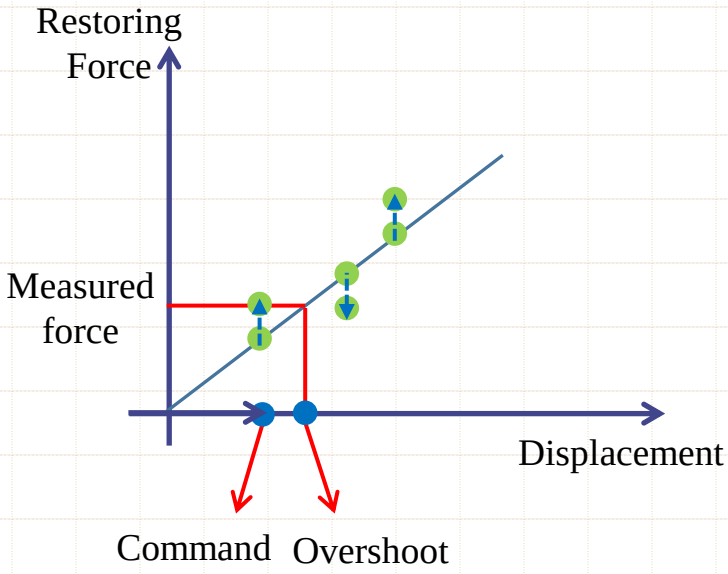
# Errors in Hybrid Simulation

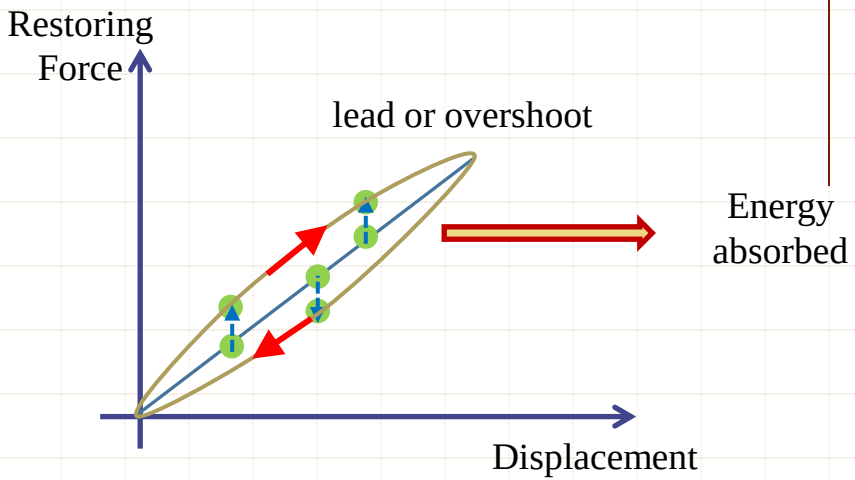
- Numerical model idealizations
  - Spatial discretization (stiffness, strength, mass distribution, modeling of energy dissipation, mesh sizes, reduced order models)
  - Time integration and equilibrium solution algorithms (time step size, algorithmic damping, numerical error propagation, instability, frequency distortion, equilibrium iterations, ill-conditioned Jacobians)
- Type of testing
  - Hold-and-ramp procedures
  - Continuous testing (avoid relaxation and stick-slip issues)
  - Rapid testing (inertia effects of test setup)
  - Real-time testing (inertia effects of test setup)

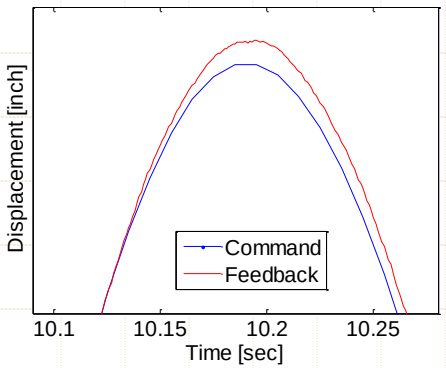
# Errors in Hybrid Simulation

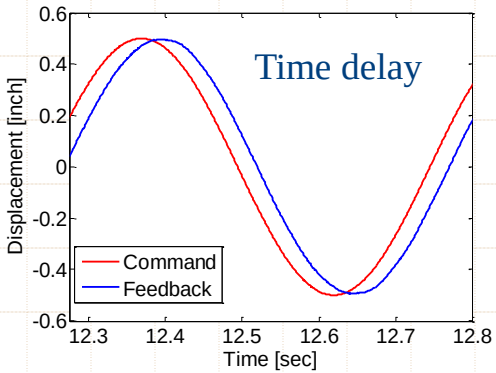
- Transfer/actuation system
  - Poorly tuned or inadequate control system induces systematic errors (lag, lead, undershoot, overshoot, feedback-loop instabilities)
  - Properties of transfer system (oil-column stiffness, cable stiffness, backlash, friction, power supply limitations, …)
  - Interaction with reaction/support structure that is too flexible
- Instrumentation and data acquisition system
  - Instrument resolution
  - Noise in feedback measurements
  - Calibration errors
  - Friction and backlash
  - A/D and D/A conversions (delays)
  - Sampling rates and filtering

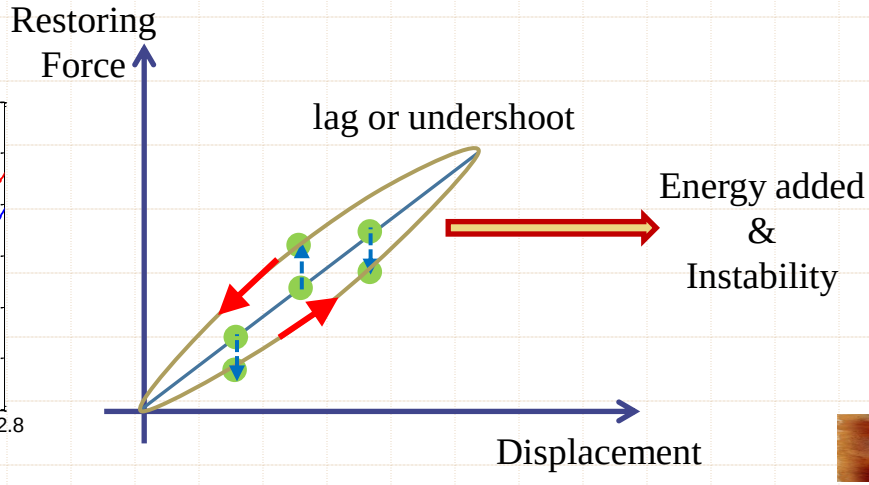
# Instability: Lag & Lead Errors



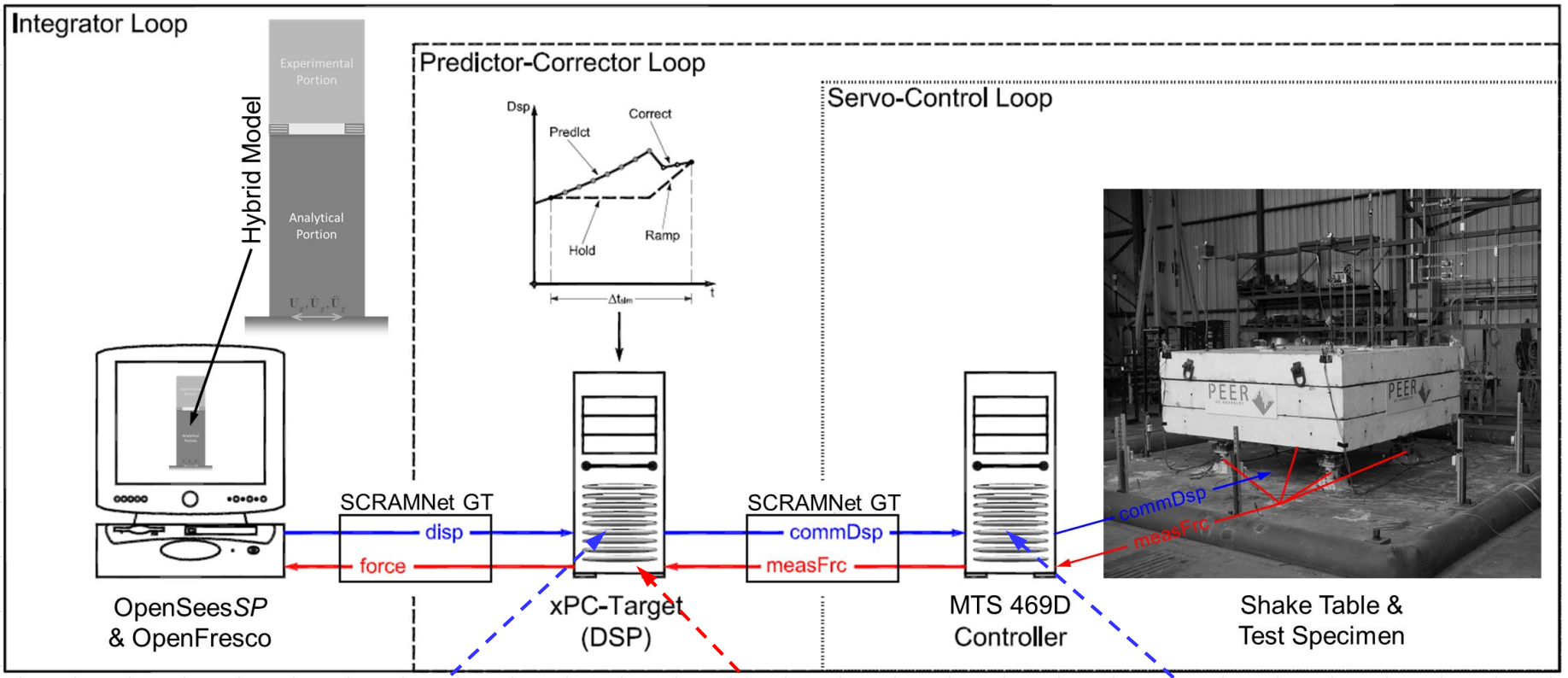



# Improving Stability & Accuracy

- Delay compensation is essential for real-time hybrid simulations (RTHS)
  - Feedforward control
  - Polynomial predictions
  - Inverse model compensation
  - Adaptive Time Series (ATS) delay compensator (works well in displacement control but not so well in force control)
- Improved control
  - Stabilization and loop-shaping
  - Nonlinear control (sliding mode control)
  - Other advanced control algorithms
- Predictor-corrector algorithms to bridge different time scales
- Force balancing for over-constrained systems
- Sensor noise reduction by filtering fbk

# Three-Loop Hardware Architecture

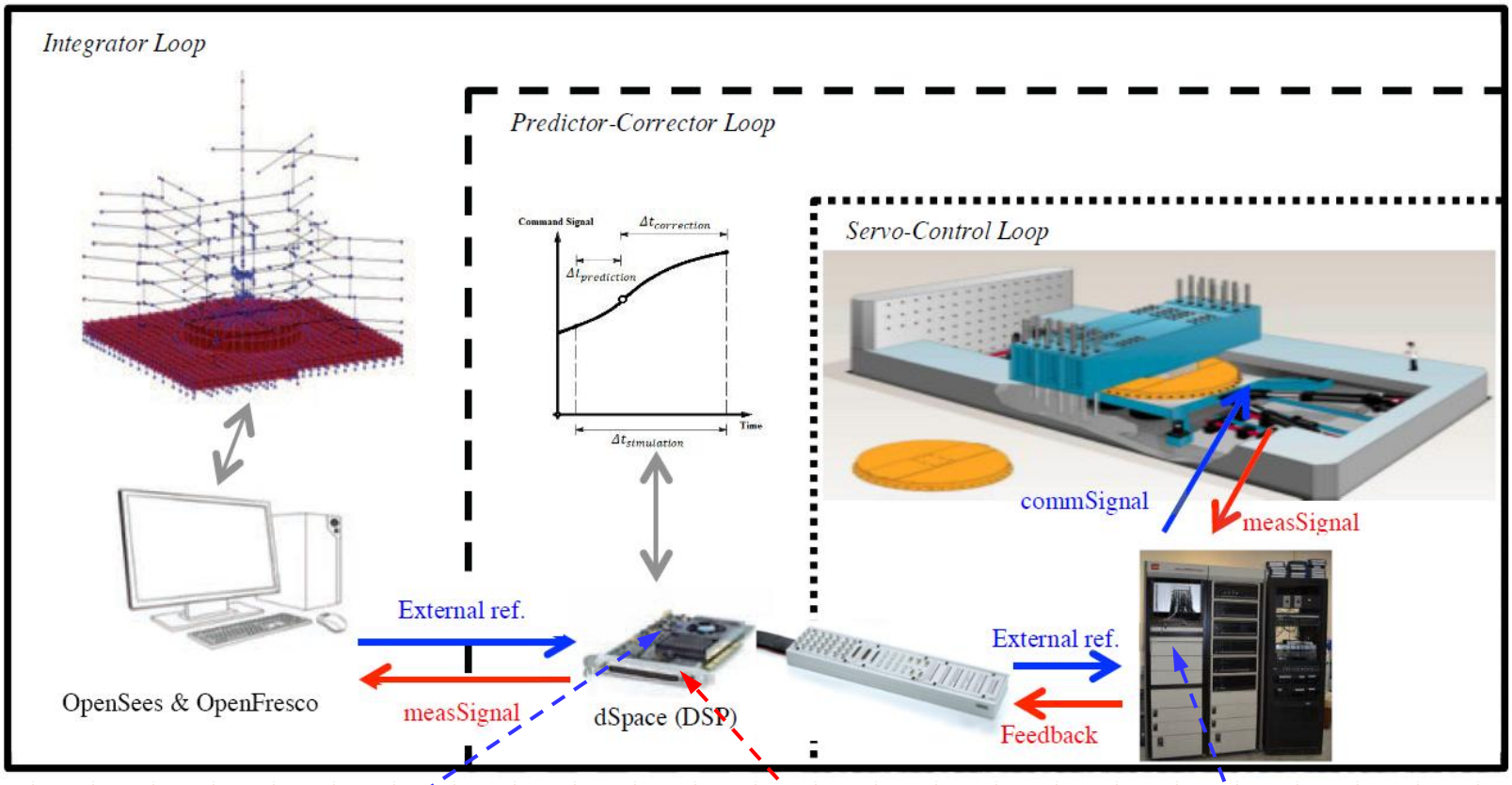



ATS delay compensator

filtering & noise reduction

TVC or other adv. ctrl. & force balancing

# Three-Loop Hardware Architecture

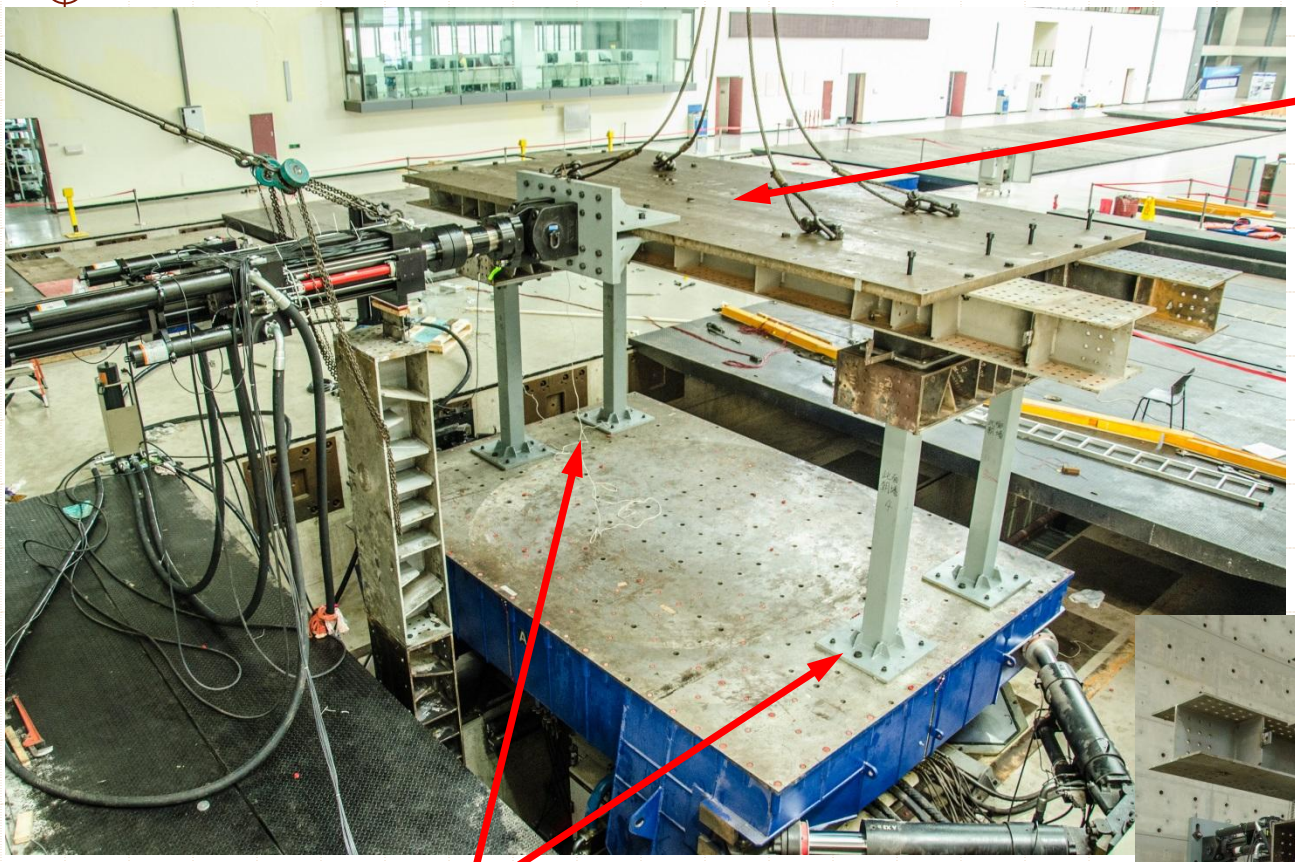



ATS delay compensator

Inertia & friction compensation, filtering & noise reduction

PID+F+Δp+ Notch ctrl.

# Using Observers for Feedback

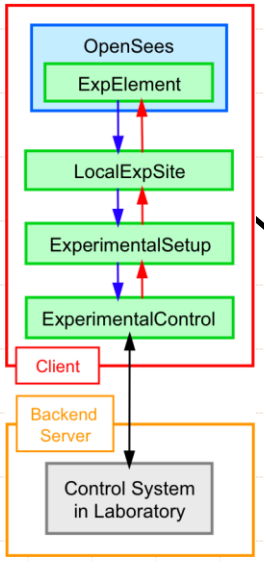


Partial-weight bridge deck

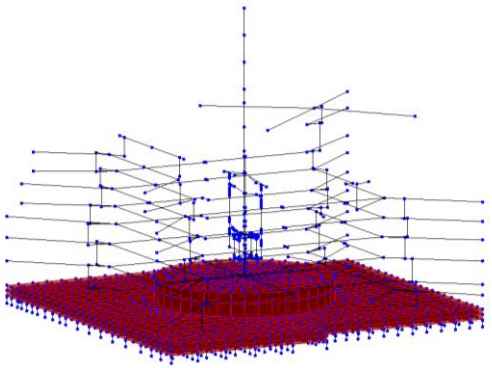


Using table observer to get shear forces at bottom of columns (load cells would be better)

# Phase1 Communication Details



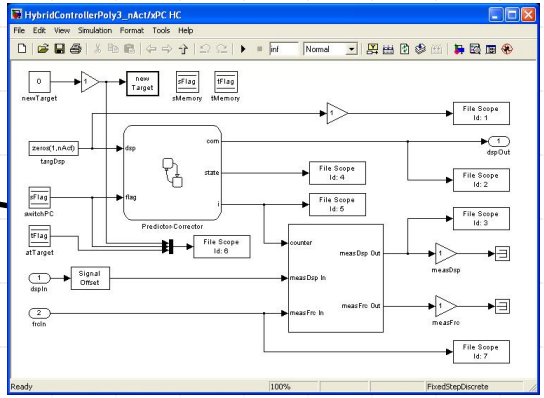

OpenSees Finite Element Model



OpenFresco Middleware

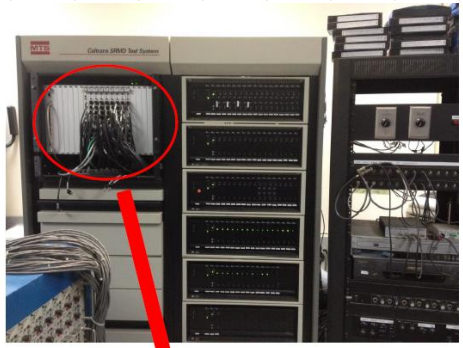
CLIB API

dSPACE real-time Predictor-Corrector

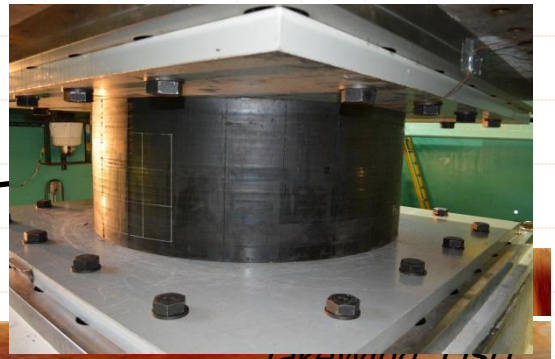


Analog BNC

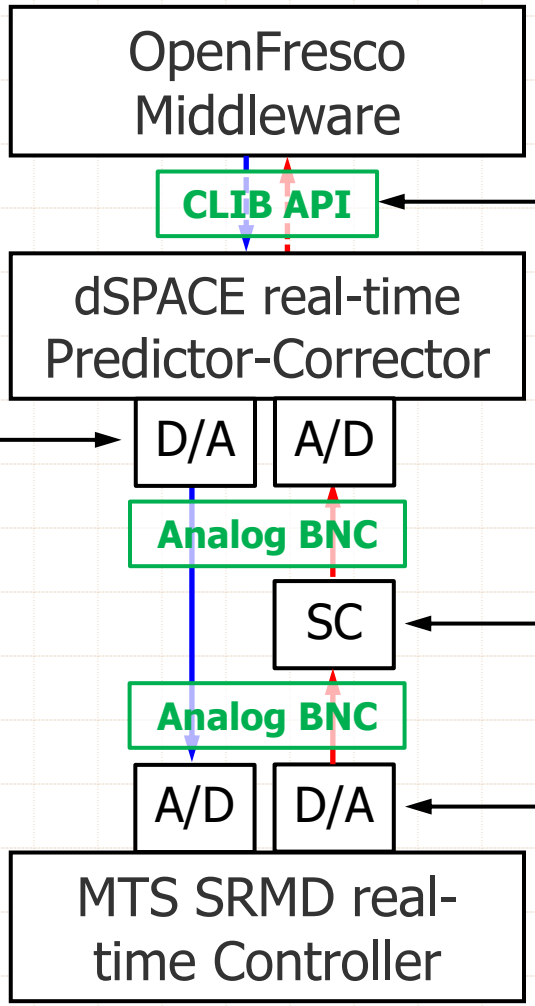
MTS SRMD real-time Controller



Physical Specimen in Laboratory



# Phase1 Communication Details

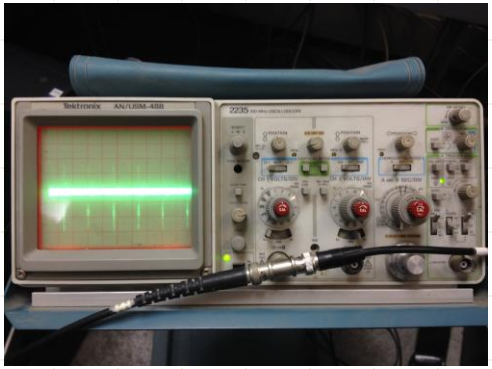



digital function calls are almost instantaneous

converters create some delay

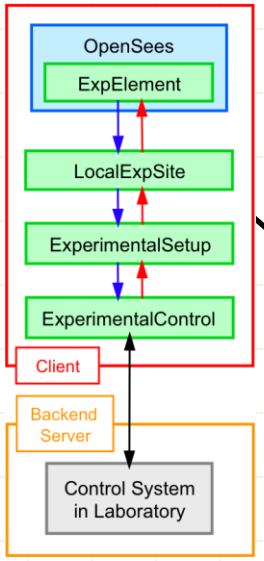
signal conditioners that filter glitches cause more delay

converters cause glitches and create some delay



glitches in SRMD fbk

# Phase2 Communication Details




OpenSees Finite Element Model

OpenFresco Middleware

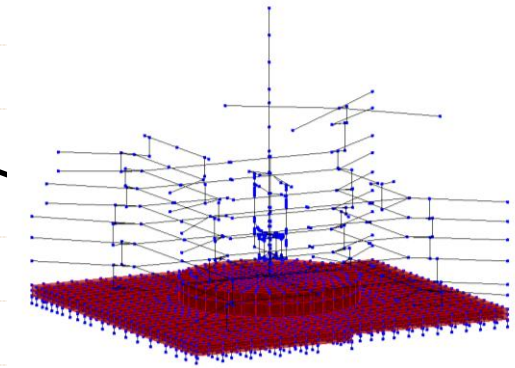
SCRAMNet+

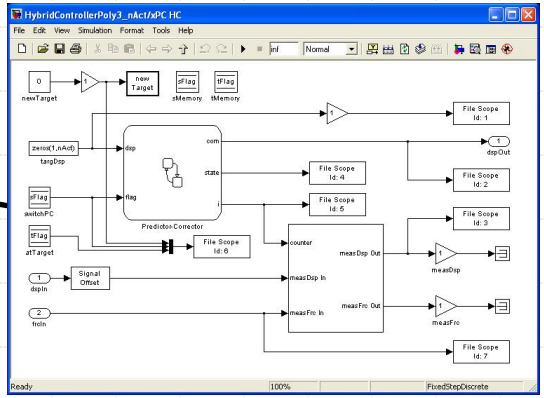
xPC-Target real-time Predictor-Corrector

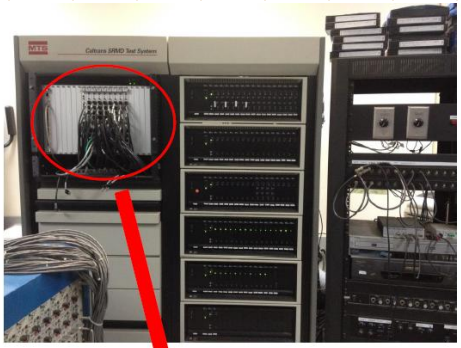
SCRAMNet+

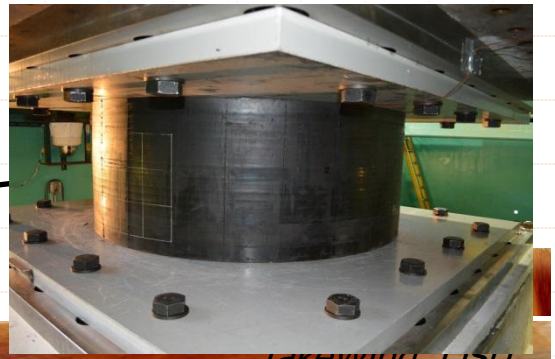
MTS SRMD real-time Controller

Physical Specimen in Laboratory

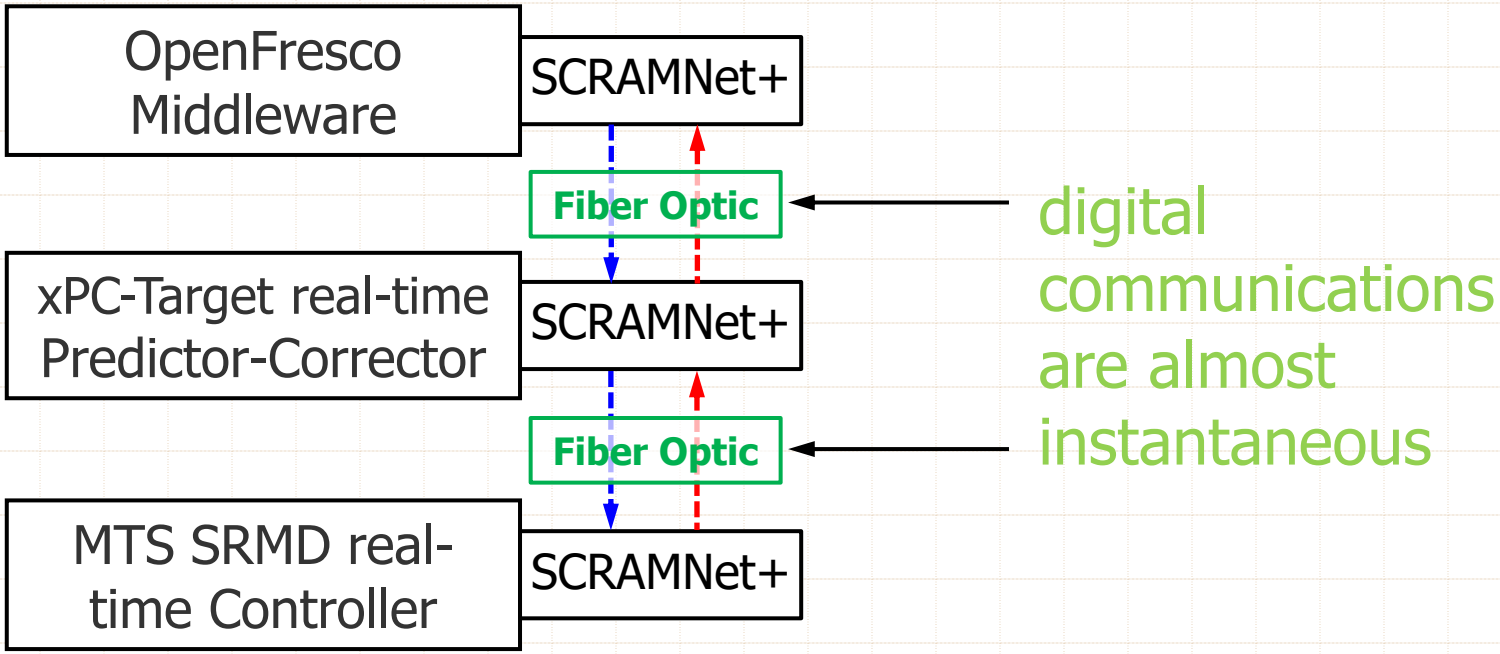






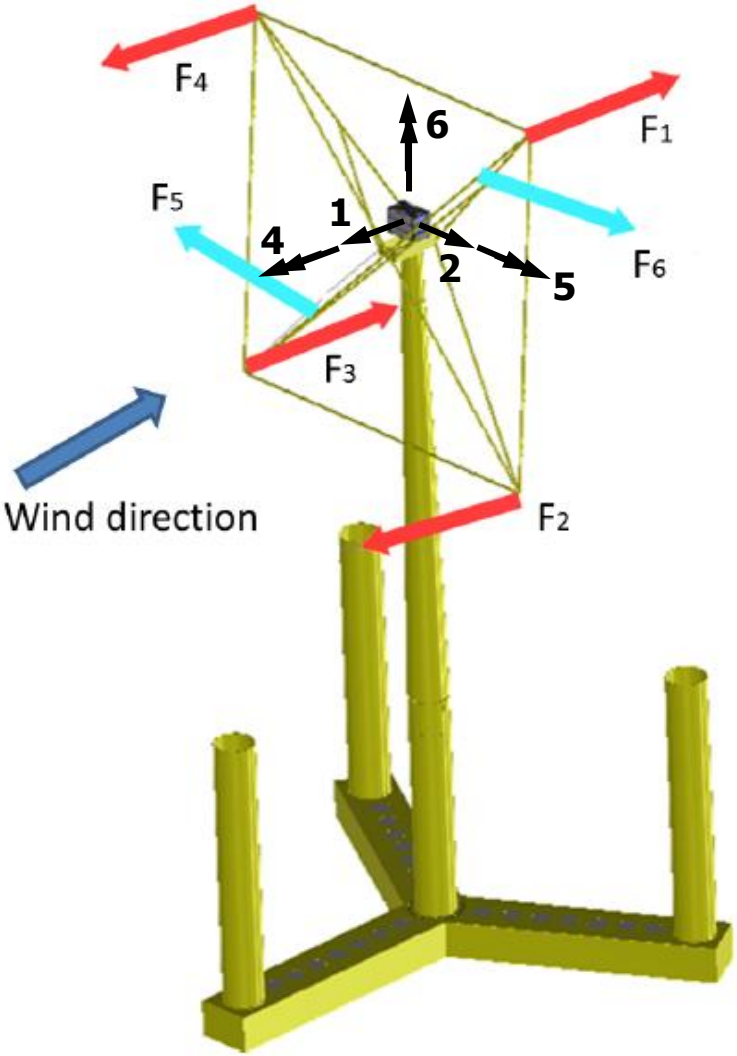


# Phase2 Communication Details




SCRAMNet : Shared Common RAM Network
[require 2 SCRAMNet+ (5V-32bit-PCI) cards and
1 SCRAMNet+ (5V-32bit-PMC) card]

# Experimental Setup



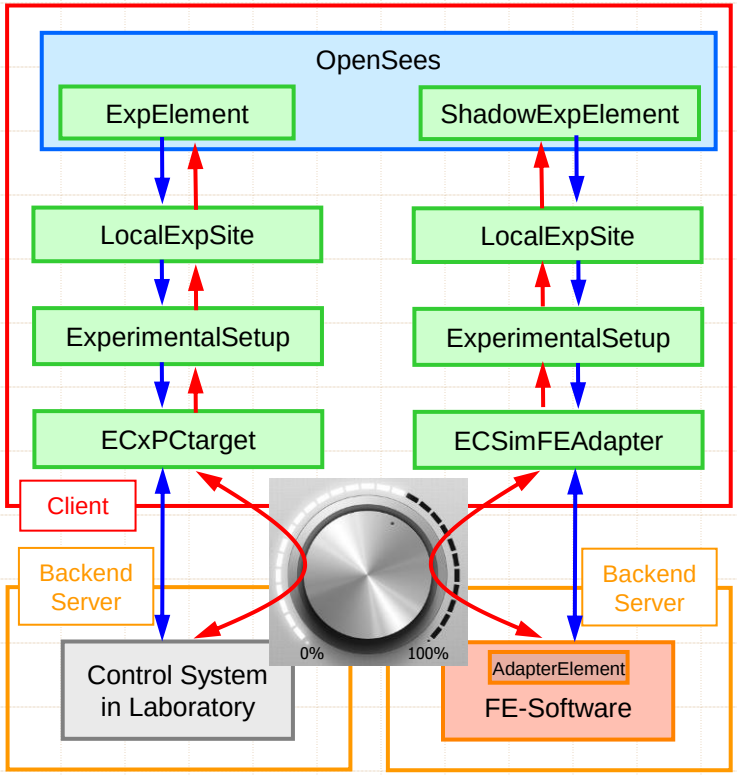

Modified from Sauder et al. (2016)

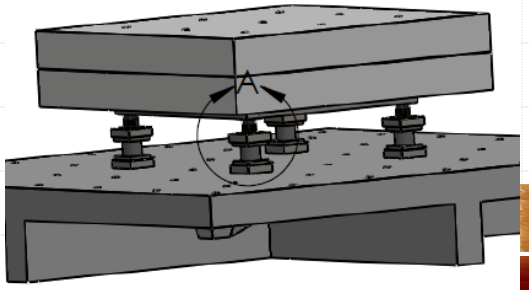
- Linear or non-linear transformations for force commands and displacement feedbacks
- Non-linear transformations can become iterative
- Account for pretension or use force balancing if system is over-constrained
- Gravity loads on test specimen always present → apply gravity loads to numerical portion before connecting with control system + apply force commands relative to start of test

# Test Rehearsal

- Use FE-Adapter element method to simultaneously connect hybrid model to a numerically simulated test specimen





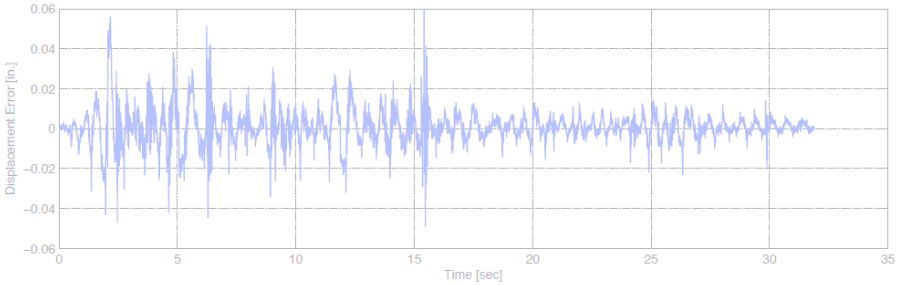



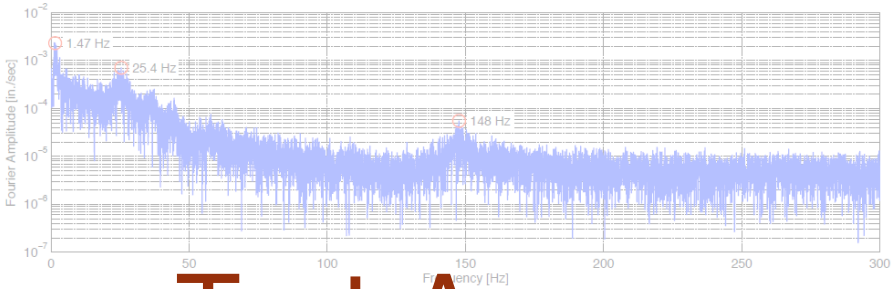
# Safety Precautions

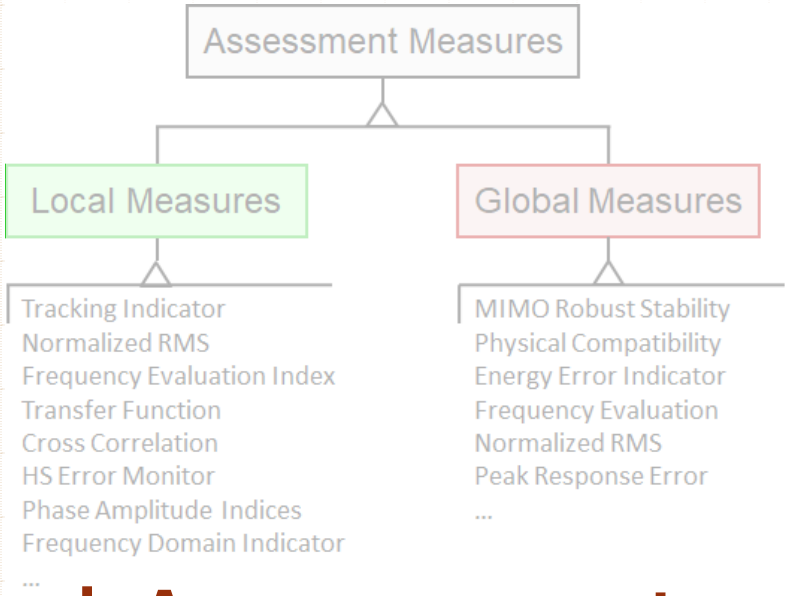
- At analysis side
  - Set limits on commands (saturation and possibly rate limit)
  - Set limit on actuator force so that once the limit is exceeded, the analysis model sends displacement commands to ramp both table and actuator to starting positions
- At controller side
  - Set both displacement and force limits so that once the limit is exceeded, the actuator pressure is switched to low, therefore, limiting the actuator force that can be applied to the specimen

# Hybrid Shake Table Test

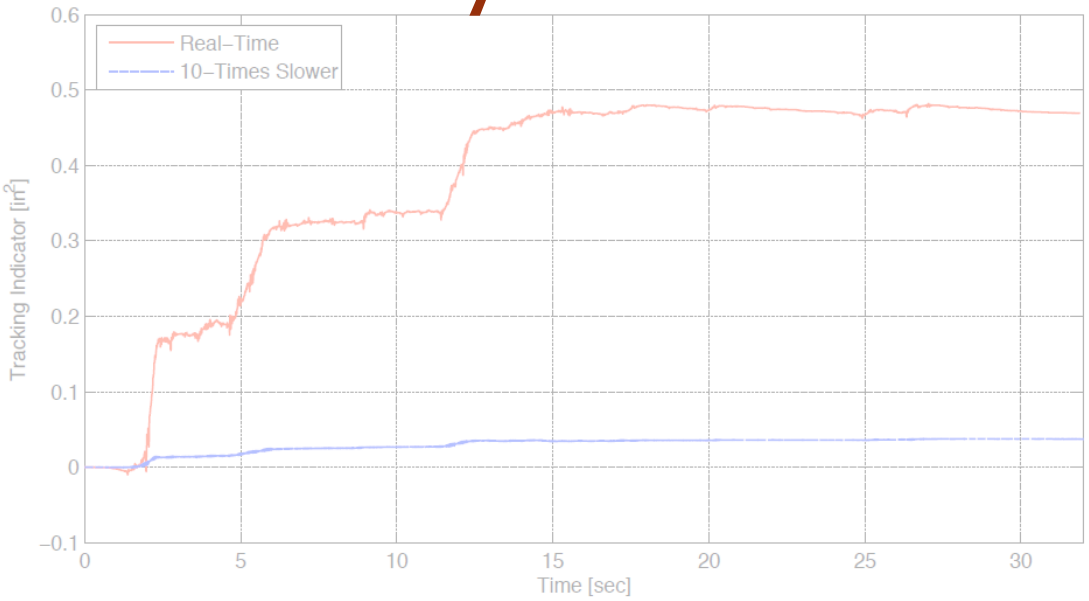




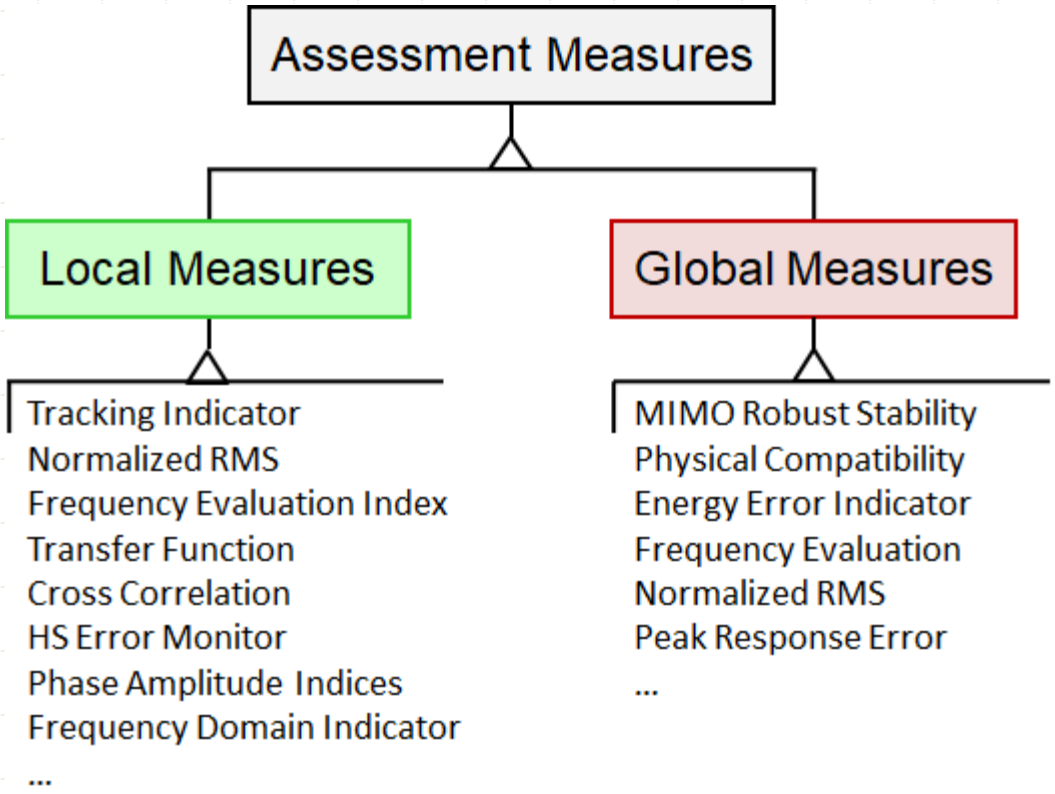



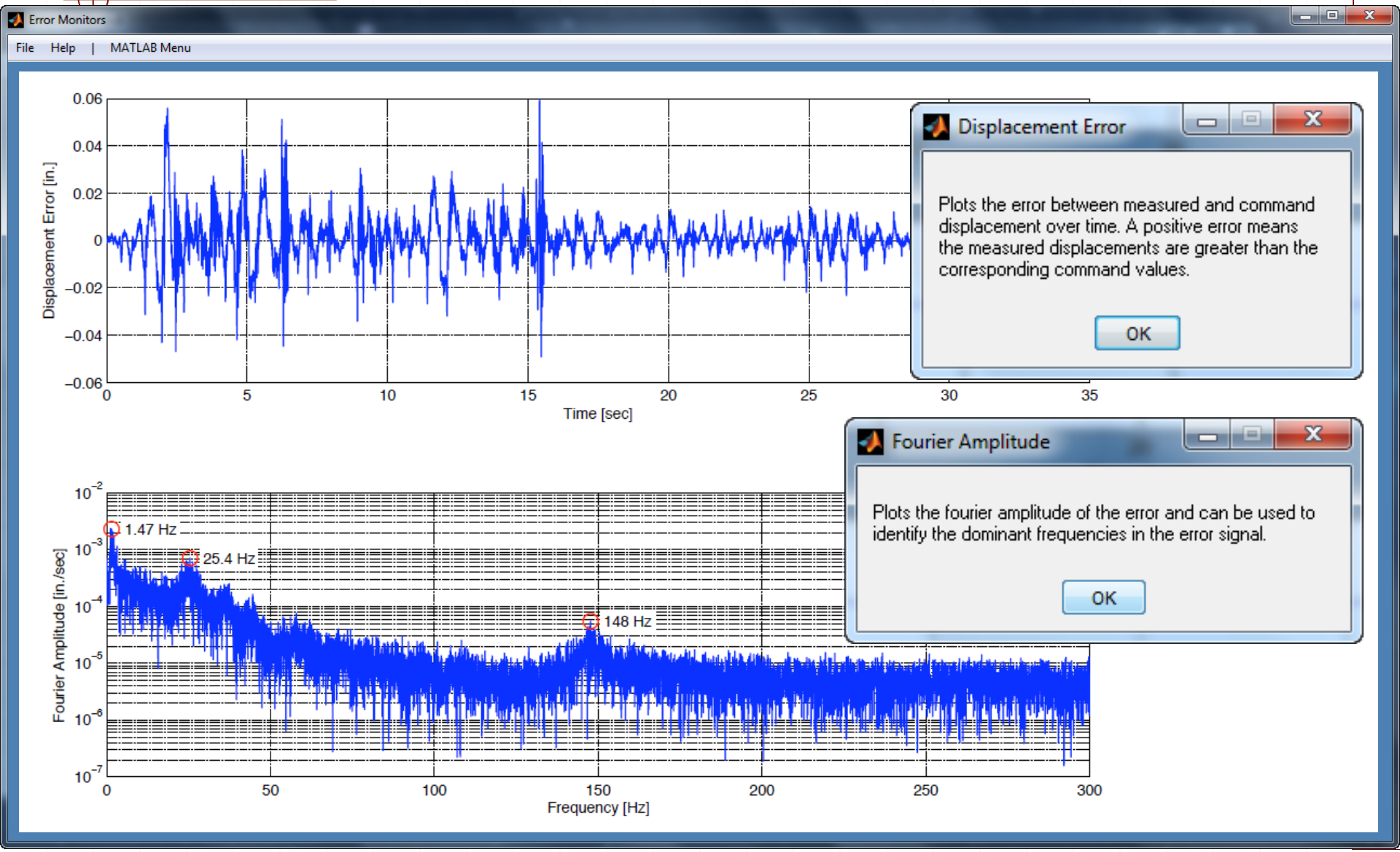

# Test Accuracy and Assessment



# Assessment Measures



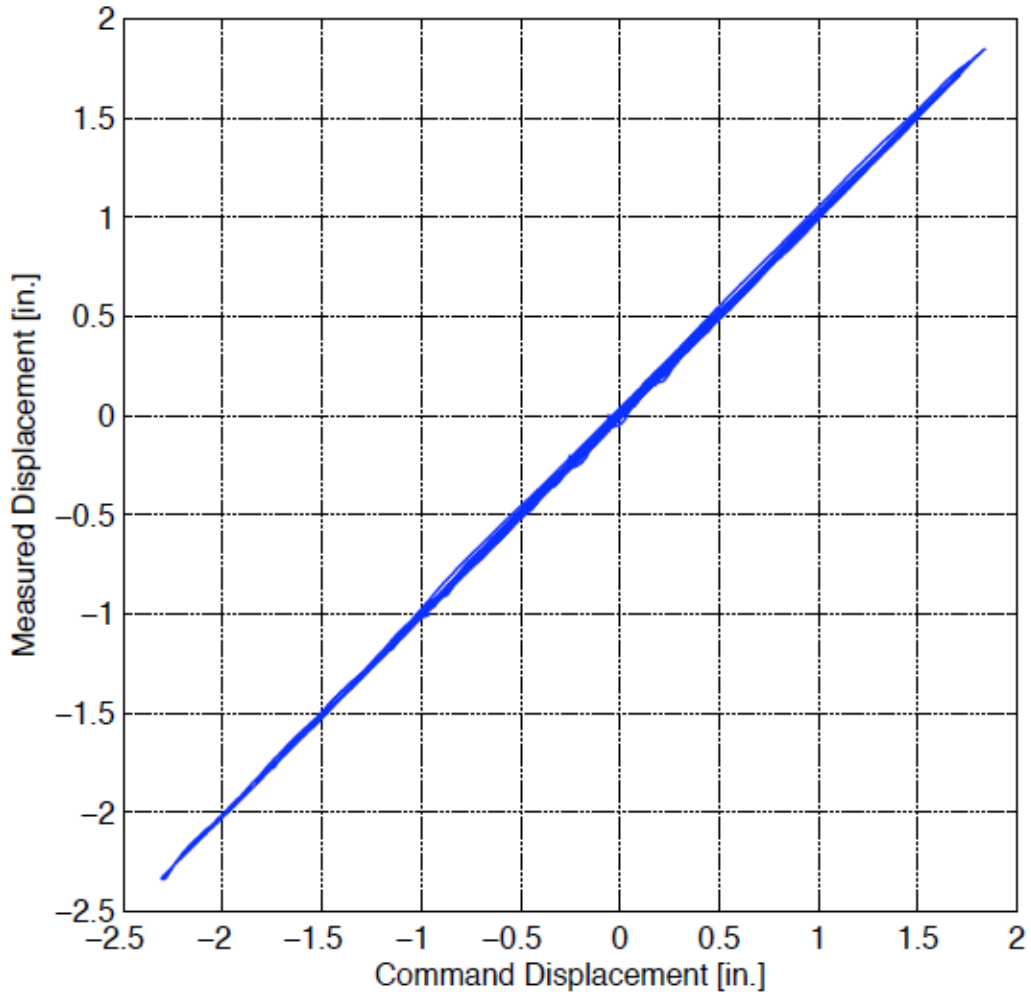
# Error Monitors: dispError and FFT



# Error Monitors: Subspace Plot

Error Monitors

File Help | MATLAB Menu

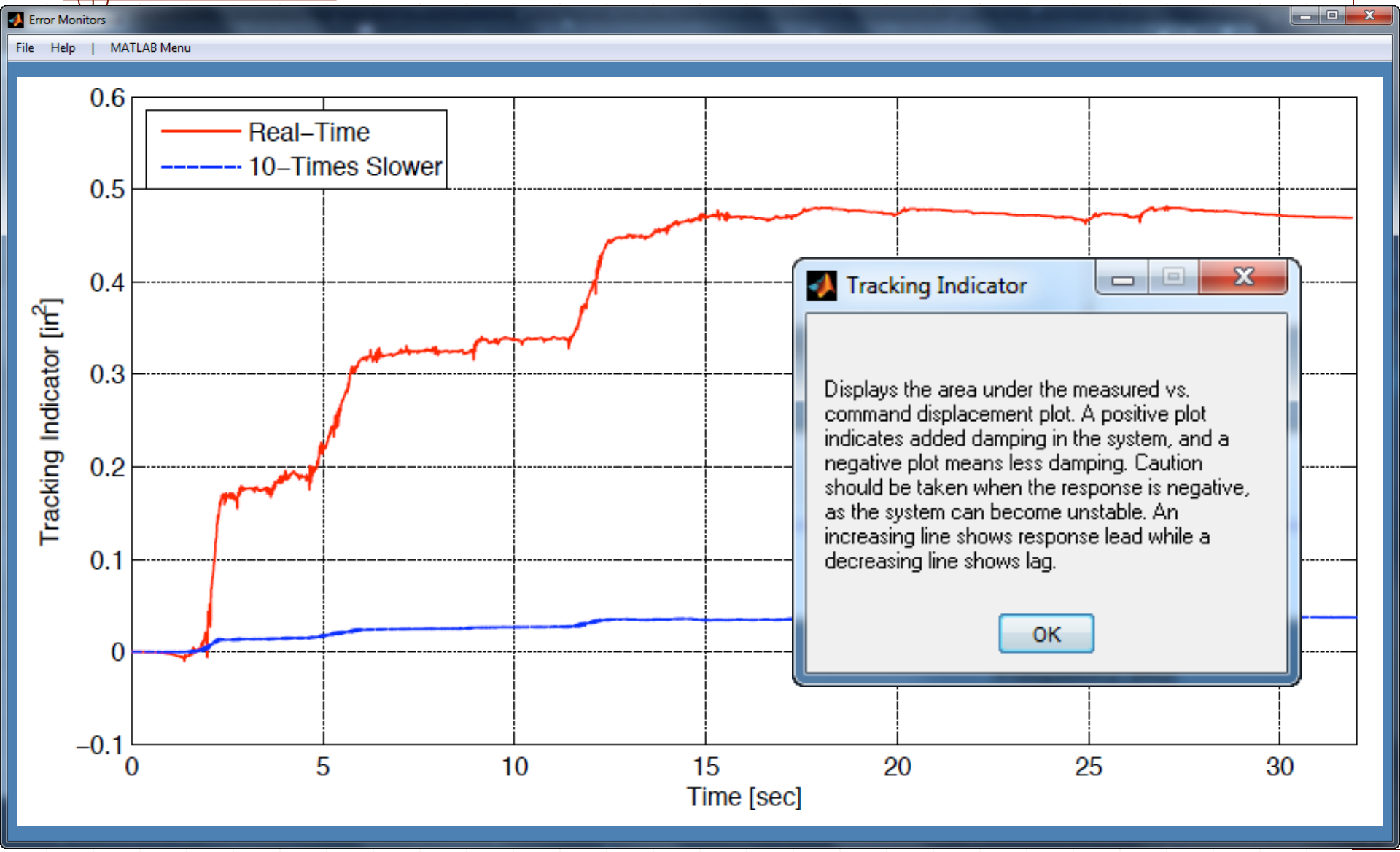



Measured v. Command...

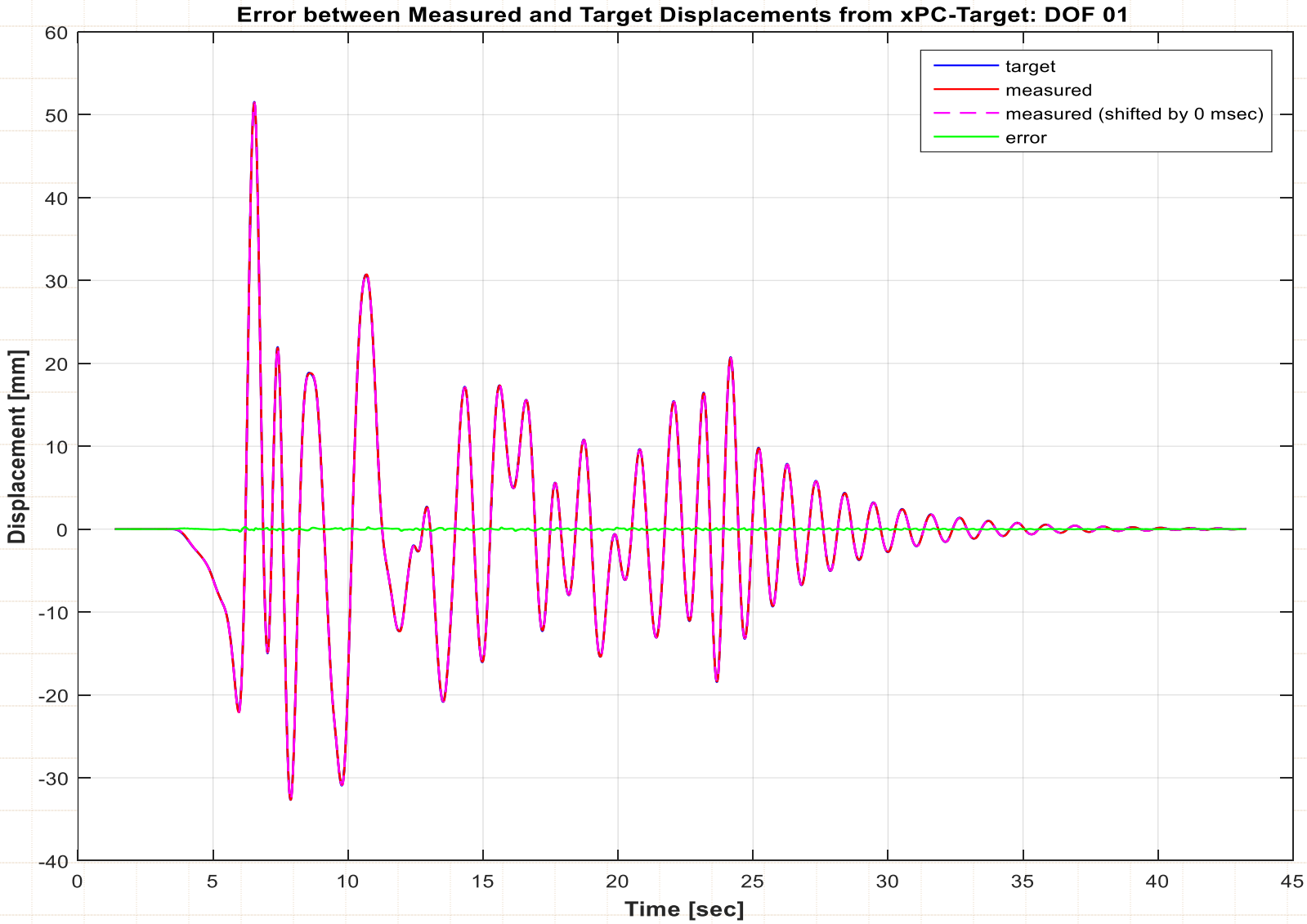
Plots measured and command displacements. A line at 45 degrees indicates perfect tracking. Less than 45 degrees means the response is undershooting, while more than 45 degrees means the response is overshooting. If the plot traces out an ellipse counter-clockwise, the response is lagging behind the command values. Similarly, a clockwise ellipse indicates a leading response.

OK

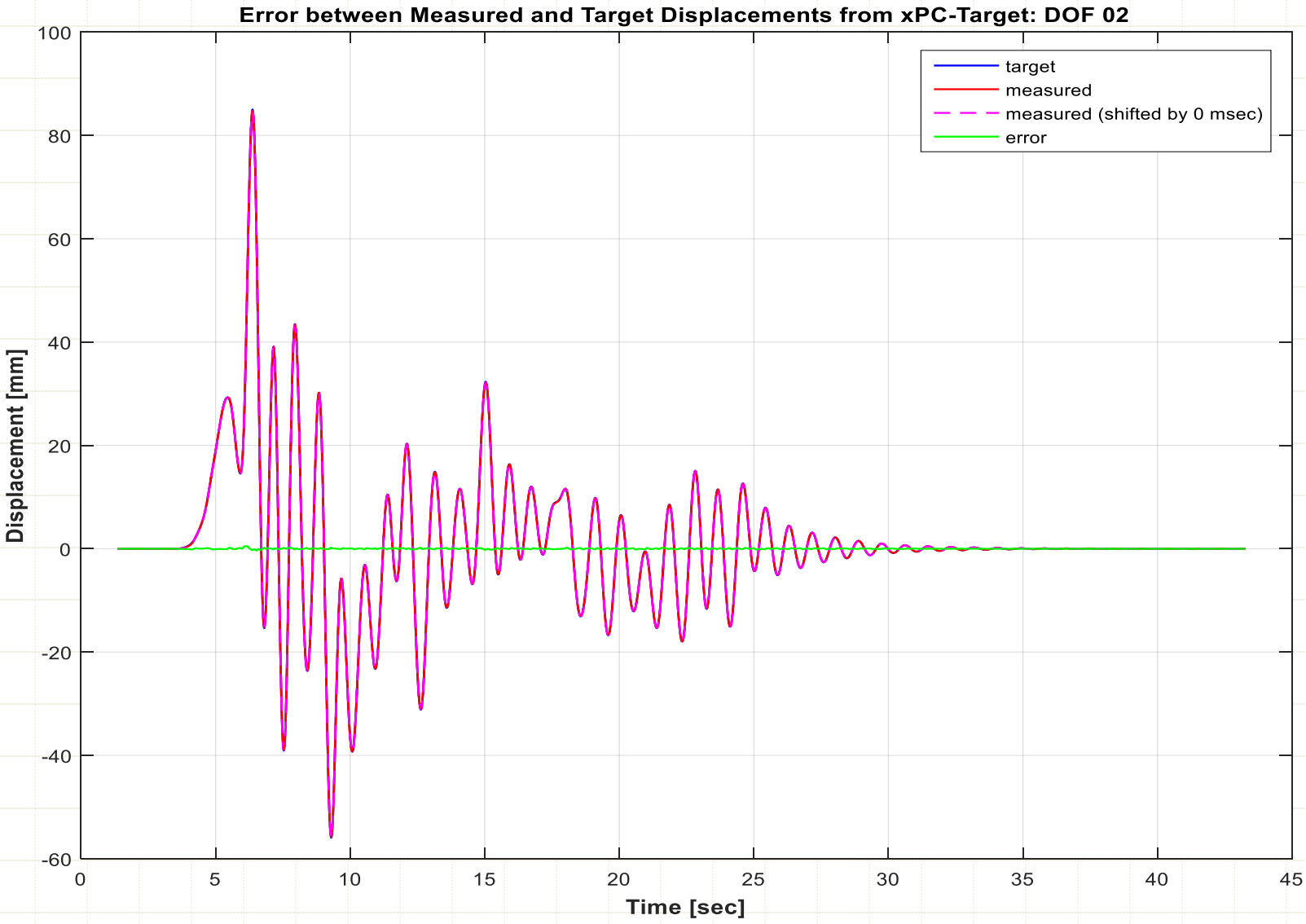
# Error Monitors: Tracking Indicator



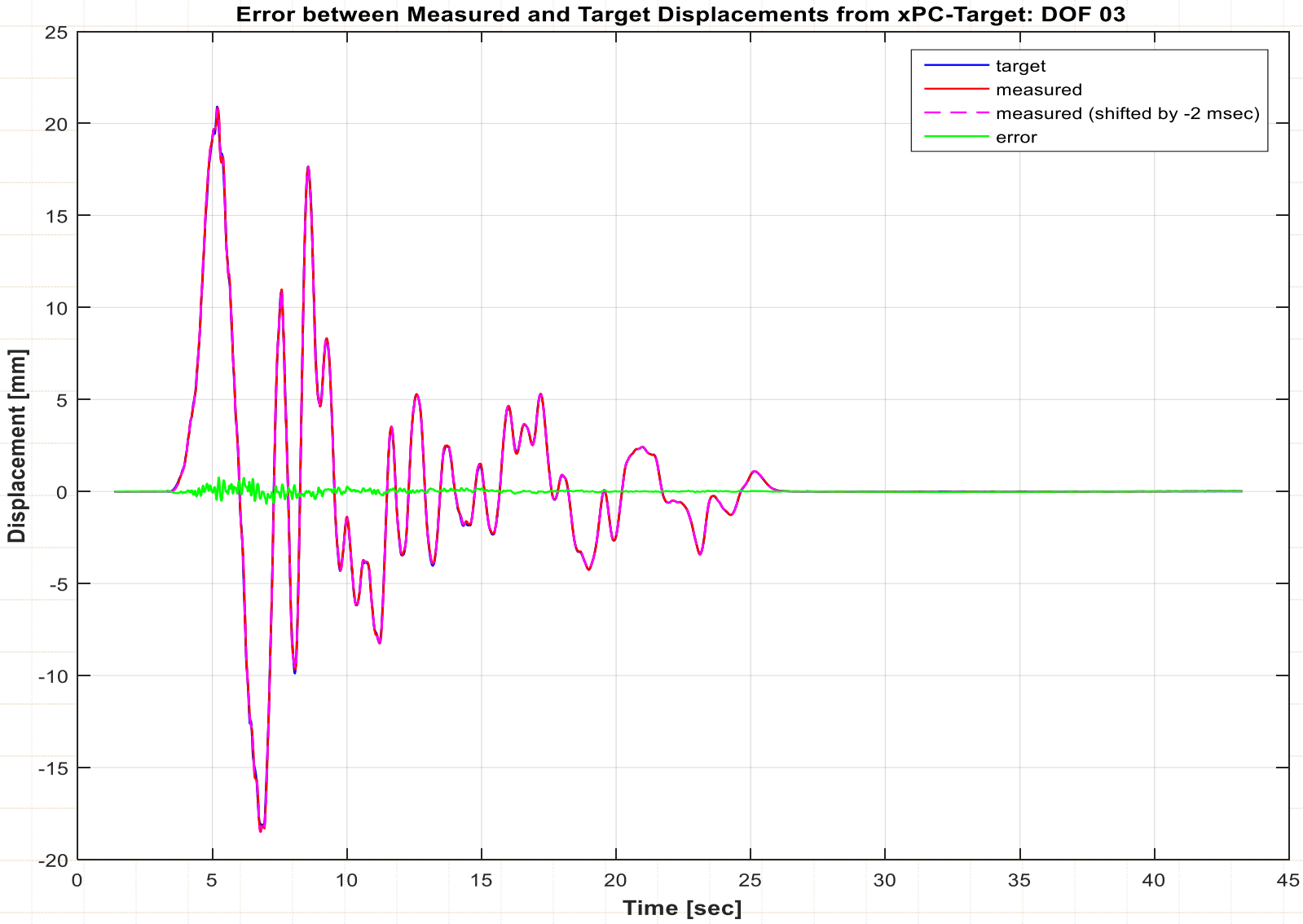
# Delay Assessment



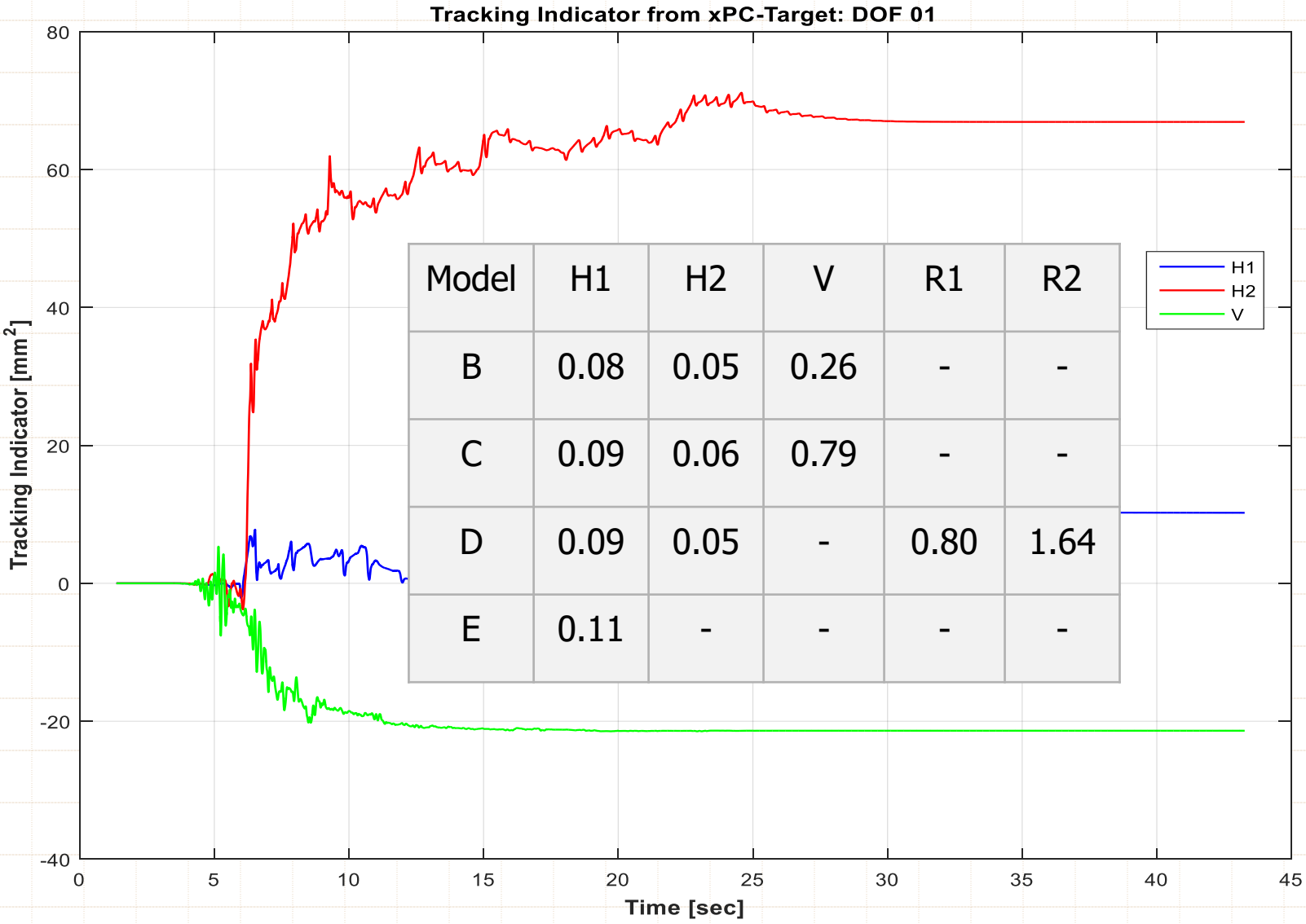
# Delay Assessment



# Delay Assessment

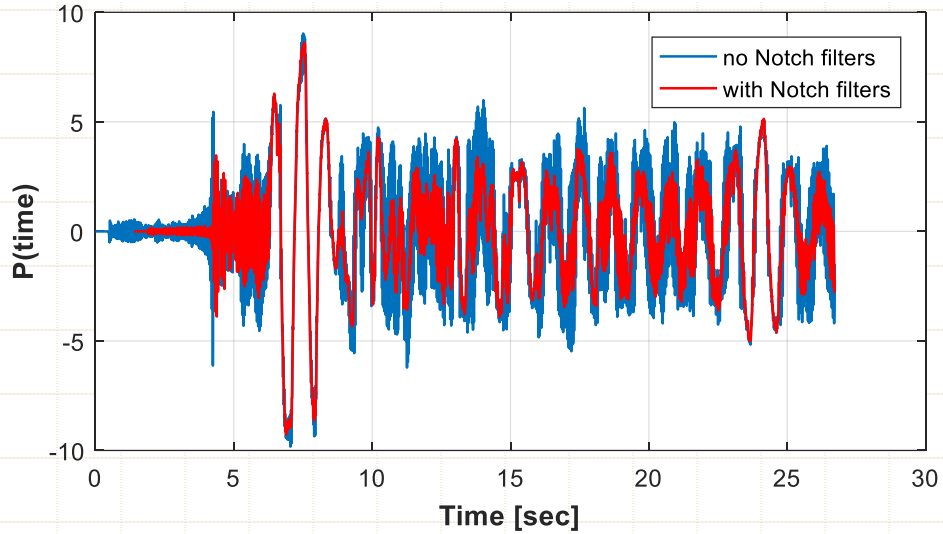


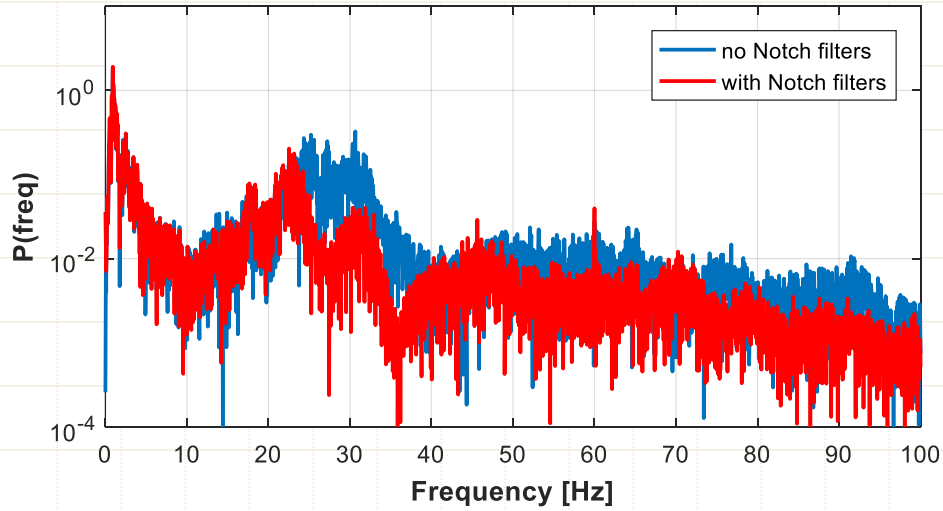
# Tracking Indicator & NRMSE[%]



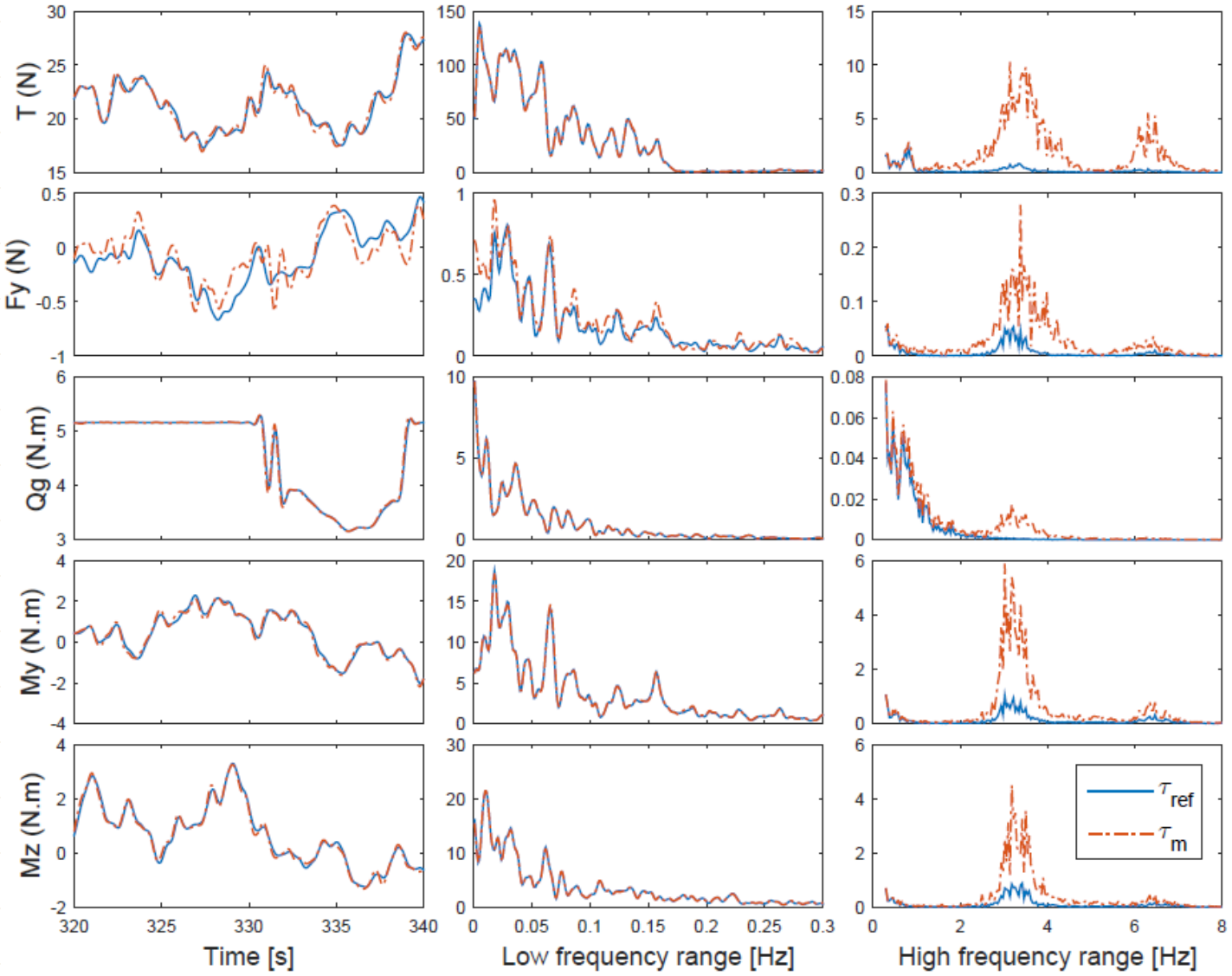

| Model | H1 | H2 | V | R1 | R2 |
|---|---|---|---|---|---|
| B | 0.08 | 0.05 | 0.26 | - | - |
| C | 0.09 | 0.06 | 0.79 | - | - |
| D | 0.09 | 0.05 | - | 0.80 | 1.64 |
| E | 0.11 | - | - | - | - |

# Filtering of Force Feedback



# Accuracy from Sauder et al. (2016)



# Summary & Conclusions

- Cyber-physical systems such as hybrid simulation testing methods are essential in moving towards improved resilience to natural hazards
- The application of hybrid simulation to wind and wave engineering problems provides many exciting research opportunities that also foster collaboration
- The OpenFresco middleware can readily be deployed and easily adapted for these new cyber-physical testing applications

# Questions?
# Thank you!





http://openfresco.berkeley.edu/

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